AirPal

Software Engineering Project

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Version 3.0

0. Table of contents

0.1. Revision History

Update number (linked)	Type of update
1	Added General overview
2	Boundaries linked with objective and updated boundaries
3	Reorganised stakeholders
4	Removed NFR references from FR2
5	Removed NFR references from FR4
6	Updated FR14, FR15,GR17, FR18
7	Details added for FR24
8	Small update on NFR4

9	Updated NFR6
10	Added NFR9 following FR4
11	
12	Added FR25
13	Added FR26
14	Added NFR10

1. Context Description

1.1 - Purpose of this document

This document provides an analysis of a system designed to manage the complexity of a modern airport ecosystem.
This document aims to:

- Present the project objectives
- Define the scope (boundaries)
- List the stakeholders involved in the system
- Define the functional requirements
- Define the non-functional requirements

1.2 - General Overview

This Project focuses on the design of a centralised management platform intended to orchestrate the multi-layered operations of a small scale airport. This system is engineered to serve as "the operational brain" of the airport, integrating a diverse workflow into a single unified digital environment to achieve maximum terminal throughput and minimise operational friction.
The core of this system revolves around "the Turnaround Cycle" which encompasses all of the significant events including but not limited to aircrafts arrival, docking, refueling, loading and departure. To simplify this complex lifecycle, the platform offers the following benefits:

Operational Management: Tools for ground-in-charge people to coordinate gates (**OB4**), landing/takeoff and real-time aircraft status (**OB1**).

Passenger Services: A digital interface for passengers to search, book, and manage tickets, including online check-in and other management facilities of the booked flight (**OB2**)

Logistics & Resources: A backend for cargo tracking (**OB3**) and the assignment of flight crews and ground staff (**OB5**)

The application runs in a real-time data environment (OB6) integrating external weather telemetry data and payment gateways that ensures a secure and scalable solution for all airport stakeholders.

1.3 - Objectives

- ### OB1 - Aircraft and fleet management

  Maintain an up-to-date aircraft inventory and schedule so that fleet availability is accurate and schedulable.

- ### OB2 - Passenger booking and self service

  Allow passengers to search and book flight tickets; manage said flights, seats and luggage; complete the check-in procedure online

- ### OB3 - Cargo and Baggage management

  Support cargo/baggage intake, basic routing and tracking, so that systems and handlers can allocate volume and trace items

- ### OB4 - Airport Operations Coordination

  Coordinate gates, terminals and flight operations to reduce gate conflicts and speed up turnarounds

- ### OB5 - Resource and staff allocation

  Provide staff rostering and allocation tools for ground staff and flight crew to ensure required coverage

- OB6 - Monitoring and operational optimization

  Provide real-time dashboards for check-in progress, cargo loading, and flight scheduling

1.4 - Boundaries

Linked Objective	IN SCOPE	OUT OF SCOPE
OB2(Booking)	Passenger flight search, booking, payment gateway integration, online check-in, digital boarding pass	Full airline revenue management or dynamic pricing engines
OB4&OB6 (Ops)	Flight status display and gate assignment dashboard	Direct Air Traffic Control system or any control of aircraft flight paths
OB1(Fleet)	Aircraft and fleet registry (availability and technical profiles) and scheduling tools	End-to-end cargo management and customs clearance
OB3 (cargo)	cargo/ baggage intake tracking (received, arrived destination)& volume allocation management	Custom logistics , legal and duty processing,real time GPS Track
OB5(Staff)	Staff rostering and role-based access for ground staff and airline users (view and assign)	Full HR and payroll for employees
OB6(monitoring)	Alerts and real-time dashboards for check-in progress, cargo loading,	Low-level hardware control on airport infrastructure,

	and flight scheduling	legal&regulatory audits
Ob6(monitoring)	Audit logs, GDPR approved data management	Legal certification, regulatory audits, compliance beyond providing features
common Ob	Integration of airline flight data, authentication service, notification service, weather service, Integration of airline booking api	Banking Transaction of internal airport transactions, Fraud detection, avionics of the cockpit,management of mechanical repairs

1.5 - Stakeholders

Human		
Stakeholder	**Type**	**Role / Interest**
Passengers	External	Search for flights and buy tickets, check-in, receive notifications
Airline staff	External	Manage flights, enter cargo manifests, view crew rosters
Airport workers	Internal	Help passengers check-in, transport and load cargo
Ground operations managers	Internal	Assign gates, monitor workers, monitor flights
Airport admin	Internal	Configure the system, manage users and permissions
Non Human		
Stakeholder	**Type**	**Role/interest**
Airlines	External	Monitoring passengers

		information, adjust ticket pricing,
Payment Gateway	External	Process ticket payments
Weather tracking service	External	Provide information about the weather outside the airport
In-house weather station	Internal	Monitor the weather on the airport grounds
Flight scheduling and status management system	Internal	Allows To modify airlines pre-scheduled flight schedules for take-off and landings of aircrafts before the flight, provide real-time flight status
Airport database	Internal	Stores all relevant user and flight data
Authentication service	Internal	Provide secure login to all users
Notification service	Internal/External	Deliver notifications and alerts

2. Functional requirements

2.1 - List of functional requirements

- ### FR1 - Registration
  - The system must allow the creation of users accounts, either by passengers for personal use or system administrators for internal workers, in order to access features reserved for authenticated users

- ### FR2 - User credentials
  - In order to create an account (see FR1), the system must ask the user to provide a Username (see NFR3), Name, Surname, email address, and a

secure password (see NFR2).

- ## FR3 - Account deletion
  - In reference to FR1, the system must allow users and system administrators to delete existing accounts (in accordance with NFR1)
- ## FR4 - Change password
  - In reference to FR1, the system must allow users to change their password.
- ## FR5 - Login
  - The system must allow unauthenticated users to authenticate, using the username and password defined in FR2
- ## FR6 - Logout
  - The system must allow authenticated users to log out of their account
- ## FR7 - Flight search
  - Following OB2, authenticated and unauthenticated users must be able to search for flights based on route and date
- ## FR8 - Seat availability management
  - Following OB1 and OB2, the system must keep track of already booked seats on all passenger flights, and display available seats on each flight displayed by FR7. The system shall restrict the number of bookable seats by value lower than the aircraft's max physical capacity for operational safety and prevent overbooking.
- ## FR9 - Ticket booking
  - Following OB2, authenticated users must be able to book tickets on passenger flights displayed by FR7
- ## FR10 - Ticket pricing management
  - Airlines must be able to update ticket pricing for each flight
- ## FR11 - Cargo shipments booking
  - Following OB3, authenticated users must be able to book cargo transportation on flights displayed by FR7

- ### FR12 - Cargo capacity management
    - Following OB3, the system must monitor and allocate available cargo capacity for each flight. The system shall maintain " a bookable threshold" that is less than the aircrafts maximum cargo capacity to account for weight limits and safety margins and display available cargo on flights displayed by FR7

- ### FR13 - Cargo pricing management
    - Airlines must be able to update cargo shipments pricing

- ### FR14 - Aircraft availability
    - Following OB1, the system must track the aircrafts' real time status.Valid states may include: at gate, maintenance, taxiing, in flight, landed.

- ### FR15 - Flight scheduling
    - Following OB1, the system must provide a scheduling interface that allows related staff to assign flight schedules to available aircrafts .

- ### FR16 - Staff assignment
    - Following OB5, the system must include a way to manually and/or automatically assign airport staff and flight crews to specific tasks/flights

- ### FR17 - Gate assignment
    - Following OB4, the system must allow Ground operations related staff to manually assign or reassign available gates to flights through a specific interface .
- ### FR18 - Landing and takeoff coordination
    - Following OB1 and OB4, the system must provide sequencing tool to manage and update and authorise landings and takeoffs based on schedules defined by FR15

- ### FR19 - Online check-in
    - Following OB2, passengers must be able to check-in to their flights online

- ### FR20 - Real-time dashboard
    - Following OB6, the system must display to authenticated users a personalized dashboard with all relevant real-time (see NFR4)

information

- ### FR21 - Alerts and notifications
  - Following OB6, the system must integrate a way to communicate important announcements and updates to users, mainly by push notifications on the users' device and emails to users' email addresses

- ### FR22 - Data storing
  - The system must be equipped of a database to store all relevant user and flight information following NFR1

- ### FR23 - Data access
  - The system must allow system admins to access information following NFR1

- ### FR24 - Weather information
  - The system must aggregate data from the In-house Weather Station (for local runway wind/visibility) and the External Weather Tracking Service (for regional flight path forecasts) to provide a bird eye view on weather conditions

- ### FR25 - boarding pass Generation
  - The system must allow system passengers to generate a boarding pass after successfully checking in Online (see FR19)

- ### FR26 - Baggage carousel finder
  - The system must allow authenticated users to search and retrieve the carousel that has been assigned for their baggages in real time (NFR4).

2.2 - Functional requirements and objectives correlation

	OB1	OB2	OB3	OB4	OB5	OB6
Functional requirements	8, 14, 15, 18	7, 8 ,9, 19	11,12	17, 18	16	20, 21

3. Non-functional requirements

3.1 - List of non-functional requirements

- ### NFR1 - Normative
  - The publicly distributed system must comply with European GDPR (Regulation 2016/679), particularly Articles 17 (Right to Erasure - FR3), 19 (Notification Obligation - FR21), and 32 (Security of processing)

- ### NFR2 - Password security
  - Following regulations specified in NFR1, the password prompted by the user during the account registration process (see FR2) must have at least 8 characters, including at least one uppercase character, one lowercase character, one digit, and one special character

- ### NFR3 - Username uniqueness
  - The username prompted by the user during the account registration process (see FR2) must be unique and cannot be accepted if another account is using the same username already

- ### NFR4 - Real-time updates
  - Flight status and gate changes must propagate to all relevant dashboards within 60s, affects FR14, FR15, FR17, FR18

- ### NFR5 - Performance
  - UI pages loading time must be < 3s, affects FR20

- ### NFR6 - Scalability
  - The system architecture must be designed to support horizontal scaling to accommodate a 500% increase in concurrent users and transactional throughput(seasonal peak loads) without exceeding the response time limits defined in NFR5

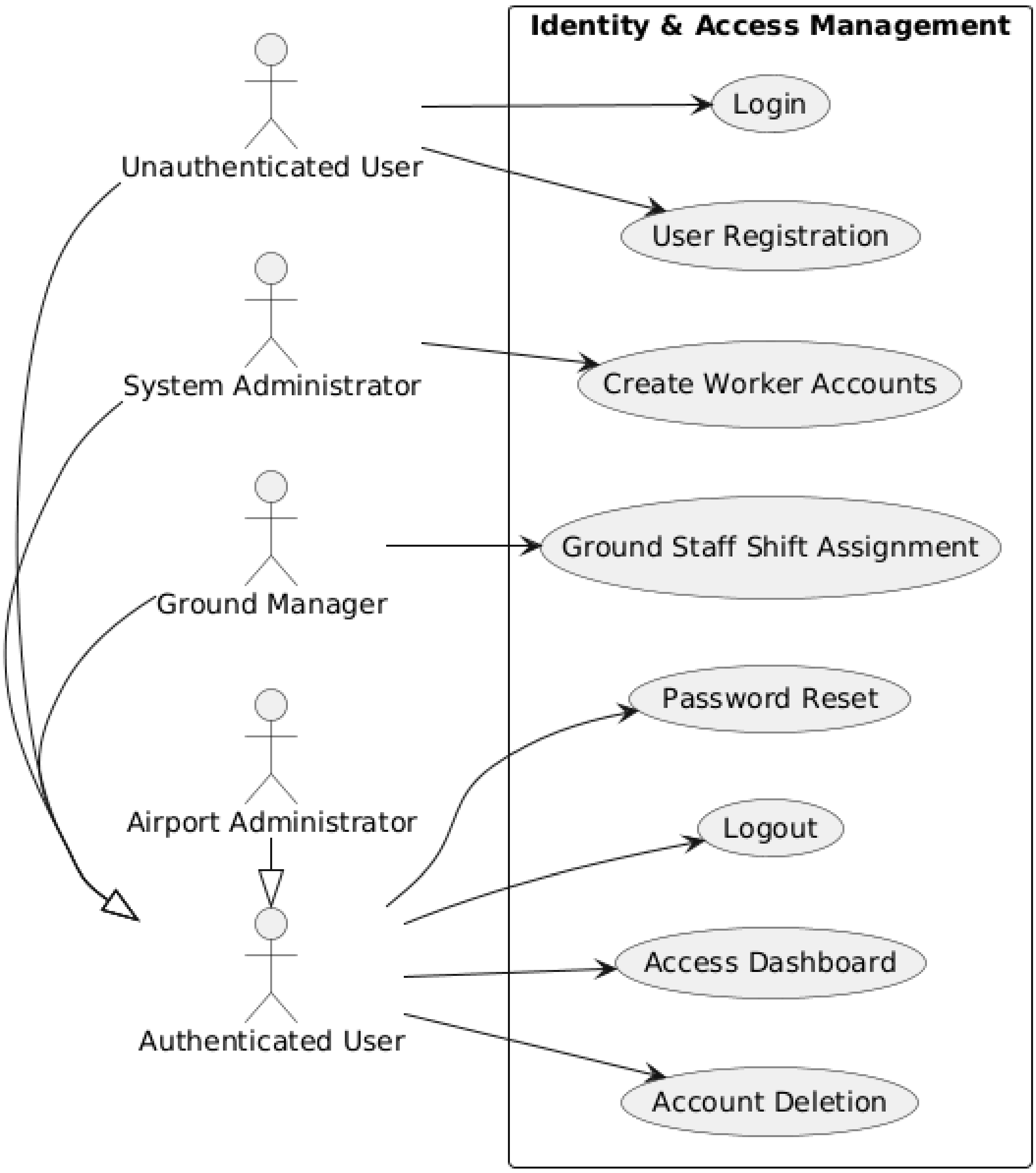
- ### NFR7 - Usability
  - The system must be intuitive, so that the users might be able to understand how to navigate the system within 10 minutes of usage without a manual

- NFR8 - Interoperability
    - The system must be able to integrate external APIs while maintaining the standards defined in NFR4, NFR5 and NFR6
- NFR9 - Secure password recovery
    - the system must allow users to change their password as stated in FR4 via a link sent directly to the user's email address as referred in.
- NFR10 - email uniqueness
    - The email provided by the user during the account registration process (see FR2) must be unique and cannot be accepted if another account is using the same email already

4. Use Cases

4.1 - Use Case diagrams

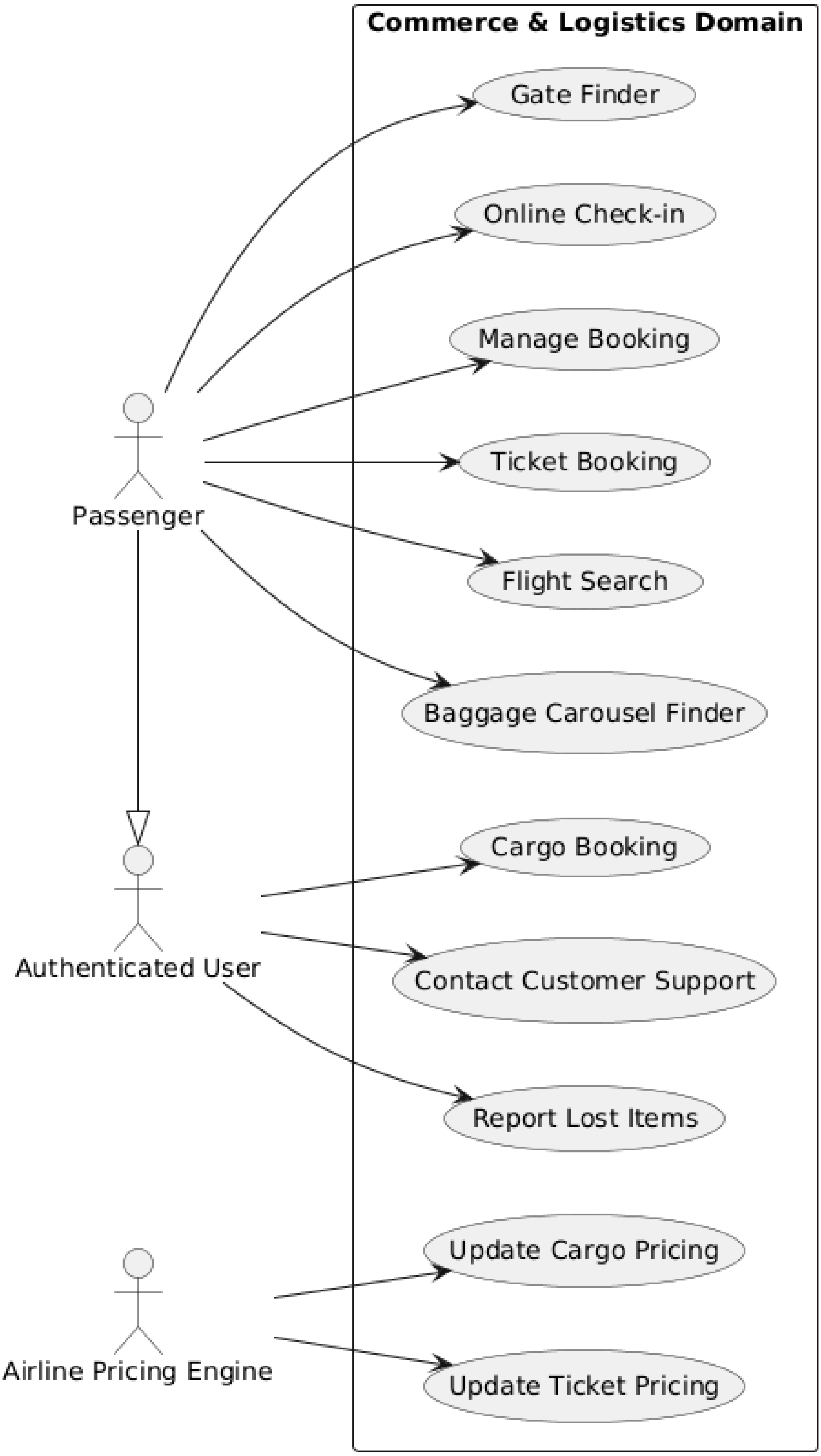
- Identity & Access Management



Use Case ID	Use Case name
UC01	User registration

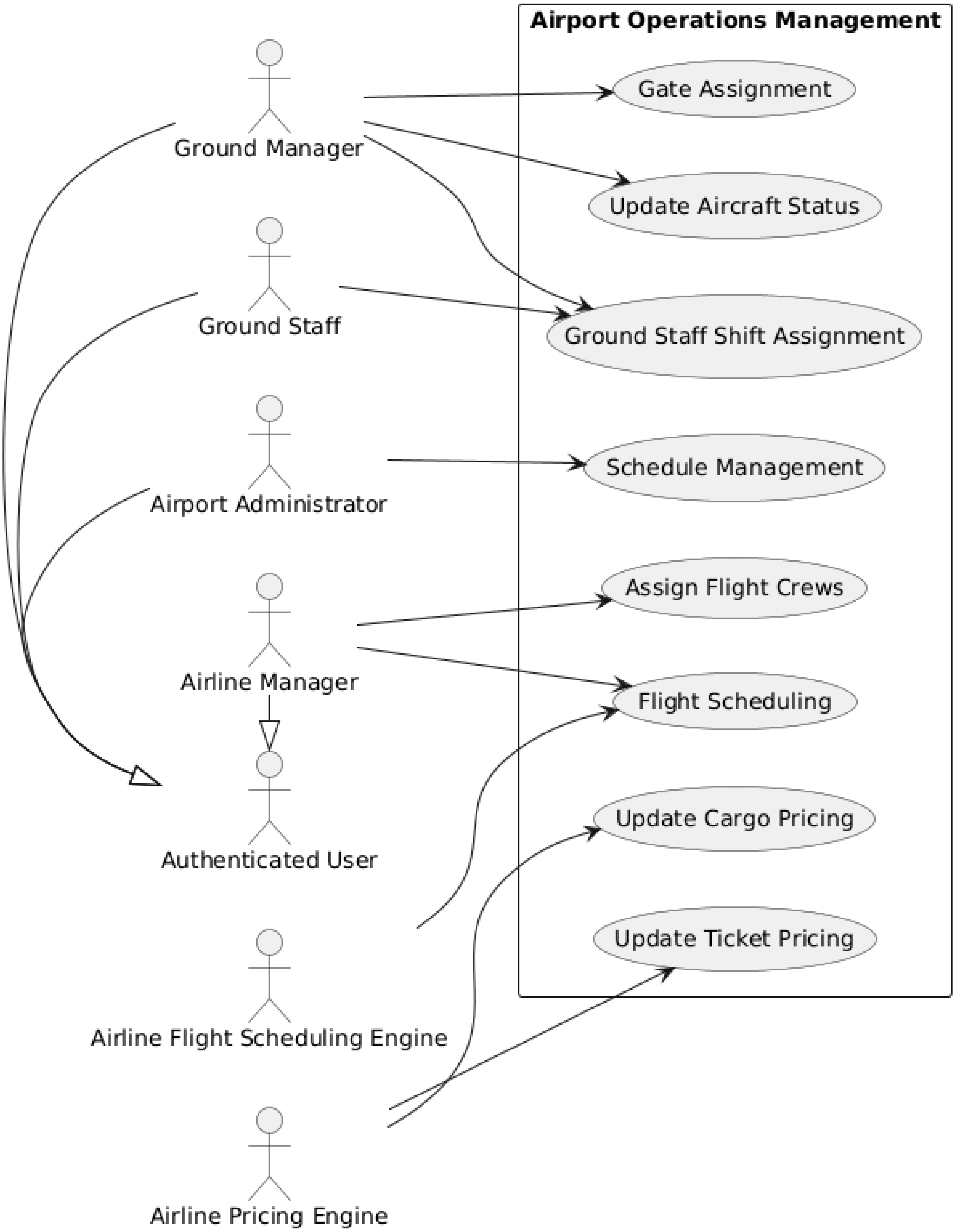
UC02	Login
UC03	Logout
UC04	Password reset
UC05	Account deletion
UC12	Access dashboard
UC24	Ground staff shift assignment
UC25	Create worker account

- Commerce & Logistics Domain



Use Case ID	Use Case name
UC06	Flight search
UC07	Ticket booking
UC08	Manage booking
UC09	Online check-in
UC10	Gate finder
UC11	Baggage carousel finder
UC13	Cargo booking
UC14	Report lost item
UC17	Customer support
UC19	Update ticket pricing
UC20	Update cargo pricing

- Airport Operations Management



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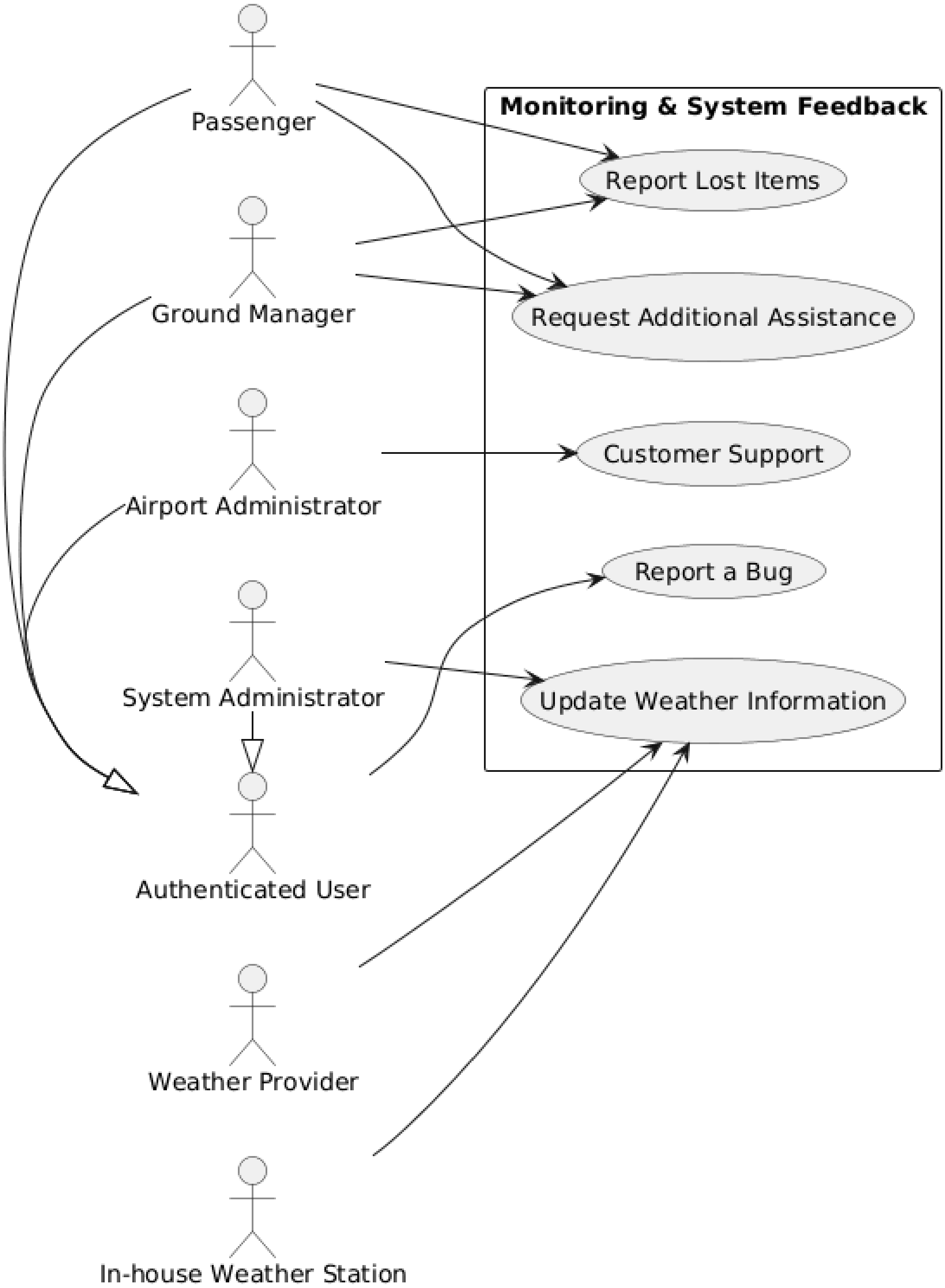
Use Case ID	Use Case name
UC018	Flight scheduling

UC21	Assign flight crews to flights
UC22	Gate assignment
UC23	Schedule management
UC24	Ground staff assignment
UC26	Update aircraft status

- ## Monitoring & System Feedback



Use Case ID	Use Case name
UC14	Report lost items
UC15	Request additional assistance for

	passengers with disabilities
UC16	Report a bug
UC17	Customer support
UC27	Update weather information

4.2 - Use Case descriptions

- ### UC01 - User registration

Use case name	User registration
Id	UC-01-UserRegistration
Description	An unauthenticated user creates a new user profile
Primary actors	Unauthenticated users
Secondary actors	Email provider
Preconditions	1. The user does not already authenticated
Trigger	An unauthenticated user wants to create an account on the platform
Main flow	1. The user clicks the “Register” button 2. The system asks the user for all the details and personal information that are needed to make a new account 3. The user inputs all the required information 4. The system verifies that the nickname and email provided by the user have not been already used in other accounts by checking with the database, and that all fields match the requirements 5. The system creates a new inactive customer account with all the information provided in the database 6. The system sends a verification link to the user’s email through the email provider

		7. The system displays a "Verify your email address" message 8. The user verifies their email address 9. The system activates the previously created customer account 10. The system displays a success message 11. The system automatically logs in the user into their account
Postconditions	Success	A new user profile is created
	Failure	An error occurred and the new user profile is not created
Alternate flows		3a. Missing information 1. The user tries to submit without filling in all the needed fields 2. The system highlights the missing fields and asks the user to input the information needed 3. Return to step 3
		4a. Invalid information 1. The information provided by the user does not follow the requirements 2. The system highlights the invalid fields and asks the user to input valid information 3. Return to step 3
		4b. Duplicate accounts 1. The system finds pre-existent user accounts that share either the nickname or the email address provided by the user 2. The system highlights the duplicate fields and displays a "Nickname or email already in use" message, and asks the user to input different information 3. Return to step 3
		5a. No database access 9a. No database access 1. The system cannot communicate with the database to save user profile data 2. The system displays an error message and prompts the user to try again 3. Return to step 2
		6a. No email provider response

	1. The system cannot communicate with the email provider 2. The system displays an error message and prompts the user to try again 3. Return to step 2

Requirement ID	Requirement name
FR1	Registration
FR2	User credentials
NFR1	Normative
NFR2	Password security
NFR3	Username uniqueness
NFR7	Usability
NFR8	Interlopability
NFR10	Email uniqueness

- ## UC02 - Login

Use case name	Login
Id	UC-02-Login
Description	An unauthenticated user logs in into their already existing account
Primary actors	Unauthenticated user
Secondary actors	none
Preconditions	1. The user has an existing account
Trigger	The unauthenticated user wants to log in into their account
Main flow	1. The user clicks the “Login” button 2. The system asks for the user’s credentials

		3. The user provides all the credentials needed 4. The system verifies that an account with those credentials exists in the database 5. The system logs in the user into their account
Postconditions	Success	The unauthenticated user has successfully logged into their pre existing account
	Failure	An error occurs and the user cannot log into their account
Alternate flows		4a. Invalid credentials 1. The system cannot find a pre existing account with the credentials provided by the user 2. The system displays an error message and prompts the user to try again with different credentials or to create a new account 3. Return to step 2
		4b. No database access 1. The system cannot communicate with the database to verify if the credentials match those of a pre existing account 2. The system displays an error message and prompts the user to try again 3. Return to step 2

Requirement ID	Requirement name
FR5	Login
NFR1	Normative
NFR7	Usability

- ### UC03 - Logout

Use case name	Logout
Id	UC-03-Logout
Description	An authenticated user logs out of their account
Primary actors	Authenticated user

Secondary actors		none
Preconditions		1. The user is already authenticated
Trigger		The user wants to log out of their account
Main flow		1. The user clicks on the "Logout" button 2. The system logs the user out of their account
Postconditions	Success	The user is successfully logged out
	Failure	none
Alternate flows		none

Requirement ID	Requirement name
FR6	Logout
NFR7	Usability

- ## UC04 - Password reset

Use case name	Password reset
Id	UC-04-PasswordReset
Description	A user resets the password for their account
Primary actors	Authenticated user, Unauthenticated user
Secondary actors	Email provider
Preconditions	1. The user has an existing account 2. The user has access to the email address associated to their account
Trigger	The user wants to reset their account's password
Main flow	1. The user clicks on the "Reset password" button 2. If the user is authenticated, the system sends a link to change the password to the user's email through the email provider

		3. The user accesses the page via the link 4. The system asks the user for a new valid password 5. The user enters the new password 6. The system saves the password in the database, updating the old one affiliated with the email address 7. The user is unauthenticated and is brought to the login page 8. Go to step 2 of UC02 main flow
Postconditions	Success	The new password is saved into the database
	Failure	An error occurs and the password does not get changed
Alternate flows		2a. Unauthenticated user 1. If the user is not authenticated, the system asks the user to input a valid email address 2. The user inputs a valid email address 3. The system searches the database for an account with the email address provided by the user 4. The system sends a link to change the password to the email address given by the user through the email provider 5. Go to step 3 of UC04 main flow
		2a.2a. Invalid email address 1. The user inputs an invalid email address 2. The system displays an error message and prompts the user to try again 3. Return to step 2a.2
		2a.3a. No database access 1. The system cannot communicate with the database to check the email address 2. The system displays an error message and prompts the user to try again 3. Return to step 2a.2
		2a.3b. Inexistent account 1. The system cannot find an account with the provided email address 2. The system displays an error message and prompts the user to try again 3. Return to step 2a.2

	2a.4a. No email provider response 1. The system cannot communicate with the email provider 2. The system displays an error message and prompts the user to try again 3. Return to step 2a.2
	2b. No email provider response 1. The system cannot communicate with the email provider 2. The system displays an error message and prompts the user to try again 3. Return to step 2
	5a. Invalid password 1. The user enters an invalid password 2. The system displays an error message and prompts the user to enter a valid password 3. Return to step 4
	6a. No database access 1. The system cannot communicate with the database to save the new password 2. The system displays an error message and prompts the user to try again 3. Return to step 5

Requirement ID	Requirement name
FR4	Change password
NFR1	Normative
NFR2	Password security
NFR7	Usability
NFR9	Secure password recovery

- ### UC05 - Account deletion

Use case name	Account deletion

Id		UC-05-AccountDeletion
Description		An authenticated user deletes their account and gets disconnected
Primary actors		Authenticated user
Secondary actors		none
Preconditions		1. The user has an existing account 2. The user is authenticated 3. The user is on the dashboard/personal area
Trigger		The user wants to delete their account
Main flow		1. The user clicks the “Delete account” button 2. The system informs the user that all data related to his account will be permanently deleted and asks if the user wants to proceed 3. The user confirms 4. The system accesses the database and deletes all data related to the user’s account 5. The system logs the user out 6. The system displays a success message
Postconditions	Success	The account is deleted and the user unauthenticated
	Failure	An error occurs and the system cannot delete the account
Alternate flows		4a. No database access 1. The system cannot access the database to delete the user’s account data 2. The system displays an error message and prompts the user to try again 3. Return to step 2

Requirement ID	Requirement name
FR3	Account deletion
NFR1	Normative
NFR7	Usability

- ### UC06 - Flight search

<table>
<tr><th colspan="2">Use case name</th><th>Flight search</th></tr>
<tr><td colspan="2">Id</td><td>UC-06-FlightSearch</td></tr>
<tr><td colspan="2">Description</td><td>A user searches for flights based on date, destination, and type (passenger/cargo), and the results are flights with available passenger seating and/or available cargo capacity</td></tr>
<tr><td colspan="2">Primary actors</td><td>Unauthenticated user,
Authenticated user</td></tr>
<tr><td colspan="2">Secondary actors</td><td>none</td></tr>
<tr><td colspan="2">Preconditions</td><td>1. Flight schedules exist in the system</td></tr>
<tr><td colspan="2">Trigger</td><td>The user wants to search for flights</td></tr>
<tr><td colspan="2">Main flow</td><td>1. The user accesses the flight search interface
2. The user enters date, destination and flight type
3. The user submits the search request
4. The system searches the database for flights matching all the parameters
5. The system displays a list of available flights, showing departure/arrival times and flight duration</td></tr>
<tr><td rowspan="2">Postconditions</td><td>Success</td><td>1. A list of available flights that match the parameters is shown to the user</td></tr>
<tr><td>Failure</td><td>1. The application displays an error message</td></tr>
<tr><td colspan="2" rowspan="2">Alternate flows</td><td>4a. No flights found
1. The system finds no flights that match the input parameters
2. The system displays a message saying “No flight available” and a “Modify your search” suggestion
3. Return to step 2</td></tr>
<tr><td>3a. Missing parameters
1. The user tries to submit without filling in all</td></tr>
</table>

	the parameters 2. The system highlights the missing parameters and asks the user to input said parameters 3. Return to step 2
	1a. No page found 1. The system cannot properly load the flight search information 2. The system displays an error message
	4b. No database access 1. The system cannot search for flights in the database 2. The system displays an error message

Requirement ID	Requirement name
FR7	Flight search
NFR5	Performance
NFR7	Usability
NFR8	Interlopability

- ## UC07 - Ticket booking

Use case name	Ticket booking
Id	UC-07-TicketBooking
Description	An authenticated user selects a flight from the search results, fills in the passenger details, pays for the ticket and gets confirmation of the booking
Primary actors	Authenticated user
Secondary actors	Payment gateway, Email / Notification provider
Preconditions	1. The user is authenticated 2. The user has searched specifically for passenger flights

		3. The user has received a list of available passenger flights
Trigger		The user wants to book a flight shown on the list of their previous search
Main flow		1. The user clicks the "Book now" button for the desired flight and enters the desired number of tickets 2. The system asks the user to enter details (name, date of birth, document info, etc.), seating and luggage options for each passenger 3. The user fills in the details for everyone and confirms 4. The system accesses the database to calculate the final cost of the booking and asks for payment details 5. The user enters their payment info and clicks pay 6. The system communicates with the payment gateway which approves the transaction 7. The system reserves the tickets for the user, creates a booking reference for the tickets, adds the reference to the user's account in the database and updates the flight's available passenger capacity 8. The system displays a success message with the booking reference and sends a confirmation email to the user's email address through the external email provider
Postconditions	Success	1. The tickets are booked, the seats are marked as taken and the user receives the confirmation
	Failure	1. An error occurs, the booking is cancelled, no money is charged and the seats remain available
Alternate flows		3a. Missing passenger/seating/luggage information 1. The user tries to confirm without giving all the information needed 2. The system highlights the missing fields and asks the user to give all the details needed 3. Return to step 3

	5a. Missing payment information 1. The user tries to pay without giving all necessary payment information 2. The system highlights the missing fields and asks the user to give all the information needed 3. Return to step 2
	6a. Denied payment 1. The payment gateway denies the transaction 2. The system displays an error message and asks the user to use a different payment method 3. Return to step 5
	6b. The system cannot communicate with the payment gateway 1. The payment gateway does not respond to the system's request to process the transaction 2. The system displays an error message and prompts the user to try again 3. Return to step 5
	7a. No database access 1. The system cannot communicate with the database to log the booking reference and all relevant data 2. The system displays an error message and prompts the user to try again 3. Return to step 2
	8a. The system cannot communicate with the email provider 1. The email provider does not respond to the system's request to notify the user 2. The system displays an error message 3. Return to step 2

Requirement ID	Requirement name
FR9	Ticket booking
NFR1	Normative
NFR5	Performance

NFR7	Usability
NFR8	Interlopability

- ## UC08 - Manage booking

Use case name	Manage booking
Id	UC-08-ManageBooking
Description	A user that has booked a flight can access all of its information and if needed modify certain details about the flight such as seating, luggage, class upgrade, rescheduling and cancellation
Primary actors	Passenger
Secondary actors	Payment gateway, Email provider
Preconditions	The user is logged in and has at least one active upcoming booking
Trigger	The user wants to modify certain aspects of their booking
Main flow	1. The passenger accesses a dedicated area where all of his active bookings are displayed 2. The passenger accesses the specific booking they want to make modifications to 3. The passenger makes modifications to the booking 4. If modifications require extra fees or a refund, the system asks the passenger to give payment method information 5. The system communicates with the payment gateway which approves the transaction 6. The system applies the changes to the booking reference and saves the data in the database 7. The system displays a success message with the updated booking reference and sends a confirmation email to the passenger's email

		address through the external email provider
Postconditions	Success	The system has successfully made modifications to a specific active booking as prompted by the passenger
	Failure	An error occurs and the system does not make modifications to a specific active booking
Alternate flows		3a. Missing payment information 1. The passenger tries to pay or receive a refund without giving all necessary payment information 2. The system highlights the missing fields and asks the passenger to give all the information needed
		4a. The payment gets denied 1. The payment gateway denies the transaction 2. The system displays an error message and asks the passenger to use a different payment method
		6a. No database access 1. The system cannot communicate with the database to log the booking reference and all relevant data 2. The system displays an error message and prompts the passenger to try again 3. Return to step 3
		7a. The system cannot communicate with the email provider 1. The email provider does not respond to the system's request to notify the passenger 2. The system displays an error message

Requirement ID	Requirement name
FR8	Seat availability management
FR9	Ticket booking
NFR5	Performance
NFR7	Usability

NFR8	Interlopability

- ## UC09 - Online check-in

Use case name		Online check-in
Id		UC-09-OnlineCheckIn
Description		An authenticated user confirms their passenger and luggage details for an upcoming flight to generate their digital boarding pass
Primary actors		Passenger
Secondary actors		Email provider
Preconditions		The user is logged in and has an active booking of a flight scheduled between 24 and 2 hours from the time the user completes the check-in procedure
Trigger		The passenger wants to generate a digital boarding pass by completing the check-in procedure
Main flow		1. The passenger accesses their dedicated personal area and selects an active booking eligible for check-in procedure 2. The passenger clicks the “Check-in” button 3. The passenger verifies passenger and luggage details and travel documents and confirms 4. The system generates the digital boarding pass for the flight and communicates with the email provider to send the digital boarding pass to the passenger’s email address 5. The system displays a “Download” button that allows the passenger to directly download the digital boarding pass from the system
Postconditions	Success	A digital boarding pass is generated and the file is sent/downloaded to/by the passenger
	Failure	The procedure fails and no digital boarding pass is generated

Alternate flows	4a. Digital boarding pass generation error 1. The system cannot generate a valid digital boarding pass requested by the passenger 2. The system displays an error message and prompts the passenger to try again
	5a. The system cannot communicate with the email provider 1. The email provider does not respond to the system's request to send the digital boarding pass to the passenger 2. The system displays an error message

Requirement ID	Requirement name
FR19	Online check-in
NFR1	Normative
NFR5	Performance
NFR7	Usability

- ### UC10 - Gate finder

Use case name	Gate finder
Id	UC-10-GateFinder
Description	A user checks the app to find out which terminal and gate a specific flight is departing from, including any last-minute changes
Primary actors	Unauthenticated users, Authenticated users
Secondary actors	none
Preconditions	1. The flight is scheduled for today 2. The flight has an assigned terminal and gate
Trigger	The user looks up a flight on a dedicated gate finder section
Main flow	1. The user accesses the dedicated "Gate finder" section

		2. The user searches for the specific flight they want information on 3. The system searches through the database to find information about the gate assigned to the flight and displays it to the user
Postconditions	Success	The gate is displayed to the user
	Failure	An error occurs and the system cannot find the information
Alternate flows		2a. Non-existent flight 1. The user inputs a non-existent flight or a flight which is not scheduled for today or has not yet been assigned a terminal and a gate 2. The system displays an error message and prompts the user to try again 3. Return to step 2
		3a. No database access 1. The system cannot communicate with the database to retrieve information about the gate assigned to the flight 2. The system displays an error message and prompts the user to try again 3. Return to step 2

Requirement ID	Requirement name
FR17	Gate assignment
NFR4	Real-time updates
NFR5	Performance
NFR7	Usability

- ### UC11 - Baggage carousel finder

Use case name	Baggage carousel finder
Id	UC-11-BaggageCarouselFinder
Description	A user looks up which baggage carousel has been

		assigned to a recently arrived flight
Primary actors		Unauthenticated user, Authenticated user
Secondary actors		none
Preconditions		1. The flight is a scheduled arrival 2. The system has assigned a baggage carousel for the flight
Trigger		The user opens the Baggage carousel finder and searches for a specific flight
Main flow		1. The user accesses the dedicated “Baggage carousel finder” 2. The user searches for the specific flight they want information on 3. The system searches through the database to find information about the baggage carousel assigned to the flight and displays it to the user
Postconditions	Success	The system shows the updated (NFR4) baggage carousel number and related info within 3 seconds (NFR5)
	Failure	An error occurs and the system cannot show the baggage carousel number
Alternate flows		2a. Arrival not found 1. The user searches for a flight that is not a scheduled arrival, so it does not have an assigned baggage carousel 2. The system displays an error message and prompts the user to try a different flight

Requirement ID	Requirement name
NFR4	Real-time updates
NFR5	Performance
NFR7	Usability

- ## UC12 - Access dashboard/personal area

Use case name		Access Dashboard/Personal area
Id		UC-12-DashboardAccess
Description		The user accesses their personal area/dashboard
Primary actors		Authenticated user
Secondary actors		none
Preconditions		1. User is authenticated 2. The system can find user data in the database
Trigger		The user opens the Dashboard / Personal area
Main flow		1. The user clicks on a button to access the dashboard/personal area 2. The system displays the appropriate dashboard/personal area depending on the type of user
Postconditions	Success	The user is on their personalized dashboard/personal area
	Failure	An error occurs and the system cannot display the appropriate dashboard/personal area
Alternate flows		2a. No page access 1. The system cannot properly display the appropriate dashboard/personal area to the user 2. The system displays an error message

Requirement ID	Requirement name
FR20	Real-time dashboard
FR22	Data storing
NFR4	Real-time updates
NFR5	Performance
NFR7	Usability

- ## UC13 - Cargo booking

<table>
<tr><th colspan="2">Use case name</th><th>Cargo Booking</th></tr>
<tr><td colspan="2">Id</td><td>UC-13-CargoBooking</td></tr>
<tr><td colspan="2">Description</td><td>An authenticated user selects a flight from the search results, fills in the cargo details, pays for the cargo and receives confirmation of the booking</td></tr>
<tr><td colspan="2">Primary actors</td><td>Authenticated user</td></tr>
<tr><td colspan="2">Secondary actors</td><td>Payment gateway,
Email provider</td></tr>
<tr><td colspan="2">Preconditions</td><td>1. The user is authenticated
2. The user has searched flights specifically to ship cargo
3. The user has successfully received a list of flights with available cargo space</td></tr>
<tr><td colspan="2">Trigger</td><td>The user wants to only ship baggage without flying as a passenger</td></tr>
<tr><td colspan="2">Main flow</td><td>1. The user clicks the “Book now” button for the desired flight
2. The system asks the user to enter the details about the baggage the user wants to ship
3. The user fills in the details and confirms
4. The system accesses the database to calculate the final cost of the booking and asks for payment details
5. The user enters their payment info and clicks pay
6. The system communicates with the payment gateway which approves the transaction
7. The system reserves the cargo space for the user, creates a booking reference, adds the reference to the user’s account in the database and updates the flight’s available cargo capacity
8. The system displays a success screen with the booking reference and sends a confirmation email to the user’s email address through the external email provider</td></tr>
<tr><td>Postconditions</td><td>Success</td><td>The user has reserved and paid for cargo space on a specific flight</td></tr>
</table>

	Failure	An error occurs and the system cancels the booking
Alternate flow		3a. Missing cargo information 1. The user tries to confirm without giving all the information needed 2. The system highlights the missing fields and asks the user to give all the details needed 3. Return to step 2
		5a. Missing payment information 1. The user tries to pay without giving all necessary payment information 2. The system highlights the missing fields and asks the user to give all the information needed 3. Return to step 5
		6a. Denied payment 1. The payment gateway denies the transaction 2. The system displays an error message and asks the user to use a different payment method 3. Return to step 5
		6b. The system cannot communicate with the payment gateway 1. The payment gateway does not respond to the system's request to process the transaction 2. The system displays an error message and prompts the user to try again 3. Return to step 5
		7a. No database access 1. The system cannot communicate with the database to log the booking reference and all relevant data 2. The system displays an error message and prompts the user to try again 3. Return to step 2
		8a. The system cannot communicate with the email provider 1. The email provider does not respond to the system's request to notify the user 2. The system displays an error message 3. Return to step 2

Requirement ID	Requirement name
FR11	Cargo shipments booking
FR12	Cargo capacity management
NFR5	Performance
NFR7	Usability
NFR8	Interlopability

- ## UC14 - Report lost items

Use case name	Report lost items
Id	Uc-14-LostItems
Description	A passenger of a recent flight can report any lost items
Primary actors	Passenger
Secondary actors	Ground manager
Preconditions	1. The user is authenticated 2. The user was a passenger on a recent flight 3. The user is on the dashboard/personal area
Trigger	The user wants to report an item lost during a recent flight of which the user was a passenger
Main flow	1. The passenger accesses a dedicated area to “Report lost items” from the dashboard/personal area 2. The system asks the passenger to give information about the lost item, including which flight was the item lost on 3. The passenger enters all relevant data 4. The system creates a ticket for the lost item in the database 5. The system notifies a ground manager of the lost item report

<table>
<tr><td colspan="2"></td><td>6. The system displays a message to the passenger that their report has been successfully saved and that a member of the airport staff will contact the passenger once the item has been found</td></tr>
<tr><td rowspan="2">Postconditions</td><td>Success</td><td>The report is valid and a ground manager is notified</td></tr>
<tr><td>Failure</td><td>An error occurs and the report doesn’t go through</td></tr>
<tr><td colspan="2" rowspan="3">Alternate flows</td><td>3a. Invalid information
1. The passenger has submitted invalid information, such as a flight he wasn't a passenger of
2. The system displays an error message and prompts the passenger to check the information and try again
3. Return to step 2</td></tr>
<tr><td>4a. No database available
1. The system cannot communicate with the database to save the ticket
2. The system displays an error message and prompts the passenger to try again
3. Return to step 2</td></tr>
<tr><td>5a. No communication with a ground manager
1. The system cannot send a notification to a ground manager
2. The system displays an error message and prompts the passenger to try again
3. Return to step 2</td></tr>
</table>

Requirement ID	Requirement name
FR20	Real-time dashboard
FR21	Alerts and notifications
FR22	Data storing
NFR1	Normative
NFR7	Usability

- ## UC15 - Request additional assistance for passengers with disabilities

Use case name		Request additional assistance for passengers with disabilities
Id		UC-15-RequestAssistance
Description		A passenger requests assistance for people with disabilities
Primary actors		Passenger
Secondary actors		Ground manager, Ground staff
Preconditions		1. The user is authenticated 2. The user is a passenger or has a booking for another passenger 3. The user is on the dashboard/personal area
Trigger		The user needs additional assistance for people with disabilities
Main flow		1. The passenger clicks a “Request additional assistance” button from their dashboard/personal area 2. The system asks the passenger to specify the additional need (for example personal guide, use of a wheelchair, etc) 3. The passenger specifies their needs 4. The system notifies a ground manager of the additional needs of the passenger 5. The ground manager assigns a ground staff member to the task 6. The system notifies the ground staff member of the task assigned by the ground manager 7. The systems displays a message of confirmation that their request for additional assistance has been successfully notified to the airport staff
Postconditions	Success	A ground staff member is given the task to assist the

		requesting passenger
	Failure	An error occurs and the system cannot indirectly provide additional assistance
Alternate flows		4a. No communication with a ground manager 1. The system cannot notify a ground manager of the additional assistance needed by the user 2. The system displays an error message and prompts the passenger to try again 3. Return to step 2
		6a. No communication with a ground staff member 1. The system cannot notify the assigned ground staff member of the task 2. The system displays an error message and prompts the ground manager to try again 3. Return to step 5

Requirement ID	Requirement name
FR16	Staff assignment
FR20	Real-time dashboard
FR21	Alerts and notifications
NFR7	Usability

- ## UC16 - Report a bug

Use case name	Report a bug
Id	UC-16-ReportBug
Description	An authenticated user reports a bug of the system
Primary actors	Authenticated user
Secondary actors	System administrator
Preconditions	1. The user is authenticated 2. The user is on the dashboard/personal area

Trigger		The user wants to report a bug in the app
Main flow		1. The user clicks on a “Report a bug” button located in the dashboard/personal area 2. The system asks the user for information about the bug 3. The user enters the information 4. The system creates a ticket for a bug report in the database 5. The system notifies a system administrator of the ticket 6. The system displays a message to the user that their report has been successfully saved
Postconditions	Success	The report is saved and a system administrator is notified
	Failure	The system cannot log the report or notify a system administrator
Alternate flows		4a. No database available 1. The system cannot communicate with the database to save the ticket 2. The system displays an error message and prompts the user to try again 3. Return to step 2
		5a. No communication with a system administrator 1. The system cannot notify a system administrator 2. The system displays an error message and prompts the user to try again 3. Return to step 2

Requirement ID	Requirement name
FR20	Real-time dashboard
FR21	Alerts and notifications
FR22	Data storing
FR23	Data access
NFR1	Normative
NFR5	Performance

NFR7	Usability

- ## UC17 - Customer support

Use case name	Customer Support
Id	UC-17-CustomerSupport
Description	An authenticated user asks for customer support
Primary actors	Authenticated user
Secondary actors	Airport admin, Ground manager
Preconditions	1. The user is authenticated 2. The user is on the dashboard/personal area
Trigger	The user needs some kind of assistance or clarification and wants to contact customer support
Main flow	1. The accesses the “Assistance” section from the dashboard/personal area 2. The system displays a list of FAQ 3. If additional assistance is needed, the user clicks on a “Contact customer support” button 4. The system asks the user for information about the issue the user is having, and whether the issue is related to a functionality of the system or a matter of the airport 5. The user gives the information about the problem 6. If the issue is about the system, the system redirects the user to UC16 and stops here, otherwise continues 7. The system creates a ticket for customer support on the database 8. The system notifies an airport administrator and/or a ground manager of the ticket 9. The system displays a confirmation message to the user that their request for customer

		assistance has been successfully saved and airport staff has been successfully notified
Postconditions	Success	The user gets redirected to UC16 or the request is saved and an appropriate airport staff member is notified
	Failure	An error occurs
Alternate flows		7a. No database access 1. The system cannot communicate with the database to save the ticket 2. The system displays an error message and prompts the user to try again 3. Return to step 4
		8a. No communication with an airport staff member 1. The system cannot notify neither an airport administrator nor a ground manager 2. The system displays an error message and prompts the user to try again 3. Return to step 4

Requirement ID	Requirement name
FR20	Real-time dashboard
FR21	Alerts and notifications
FR22	Data storing
NFR1	Normative
NFR7	Usability

- ### UC18 - Flight scheduling

Use case name	Flight scheduling
Id	UC-18-FlightScheduling
Description	An airline flight scheduling engine provides the system with flight schedules

Primary actors		Airline manager
Secondary actors		Airport administrator, Ground manager, Airline flight scheduling engine
Preconditions		1. The user is authenticated and has airline manager permissions 2. The airline flight scheduling engine has created a schedule
Trigger		The airline manager wants to submit the flight schedule to the airport
Main flow		1. The airline manager puts in contact the system with the airline flight scheduling engine 2. The system receives from the airline flight scheduling system the new flight schedule 3. The system accesses the database to retrieve the airport's schedule 4. The system compares the two schedules 5. The system adds the new schedule to the database 6. The system shows a success message to the airline manager 7. The system notifies airport admins and ground managers of the new schedule added
Postconditions	Success	The new flight schedule is added to the database
	Failure	An error occurs and the new flight schedule is not added to the database
Alternate flows		2a. No communication between systems 1. The system cannot communicate with the airline flight scheduling engine 2. The system displays an error message and prompts the airline manager to try again 3. Return to 1
		3a. No database access 5a. No database access 1. The system cannot communicate with the database to read or write relevant information 2. The system displays an error message and prompts the airline manager to try again

	3. Return to 1
	4a. Incompatible flight schedules 1. The system finds incompatibilities between the old and new flight schedules 2. The system displays an error message and prompts the airline manager to generate a new schedule with the airline flight scheduling engine and try again 3. Return to 1

Requirement ID	Requirement name
FR14	Aircraft availability
FR15	Flight scheduling
FR18	Landing and takeoff coordination
NFR4	Real-time updates
NFR5	Performance
NFR6	Scalability

- ### UC19 - Update ticket pricing

Use case name	Update ticket pricing
Id	UC-19-UpdateTicketPricing
Description	The airline pricing engine updates the ticket prices for available passenger flights
Primary actors	Airline pricing engine
Secondary actors	Airport admin, Airline manager
Preconditions	1. Ticket pricing needs to be updated according to the airline pricing engine
Trigger	A change in ticket pricing needs to happen
Main flow	1. The airline pricing engine provides

		information about updated ticket prices to the system 2. The system saves the data in the database 3. The system notifies both an airport administrator and an airline manager of the successful operation
Postconditions	Success	Ticket prices are successfully updated
	Failure	An error occurs and ticket prices do not change
Alternate flows		1a. No communication between systems 1. The airline pricing engine cannot communicate with the system
		2a. No database access 1. The system cannot communicate with the database to save the new data 2. The system notifies an airport administrator and an airline manager of the error 3. Return to step 1
		2a. No communication with airport administrator or airline manager 3a. No communication with airport administrator or airline manager 1. The system cannot notify an airport administrator and/or an airline manager 2. Return to step 1

Requirement ID	Requirement name
FR10	Ticket pricing management
NFR6	Scalability
NFR8	Interlopability

- ### UC20 - Update cargo pricing

Use case name	Update cargo pricing
Id	UC-20-UpdateCargoPricing

<table>
<tr><td colspan="2">Description</td><td>The airline pricing engine updates prices for cargo space available</td></tr>
<tr><td colspan="2">Primary actors</td><td>Airline pricing engine</td></tr>
<tr><td colspan="2">Secondary actors</td><td>Airport admin,
Airline manager</td></tr>
<tr><td colspan="2">Preconditions</td><td>1. Cargo pricing needs to be updated according to the airline pricing engine</td></tr>
<tr><td colspan="2">Trigger</td><td>A change in cargo pricing needs to happen</td></tr>
<tr><td colspan="2">Main flow</td><td>1. The airline pricing engine provides information about updated cargo prices to the system
2. The system saves the data in the database
3. The system notifies both an airport administrator and an airline manager of the successful operation</td></tr>
<tr><td rowspan="2">Postconditions</td><td>Success</td><td>Cargo prices are successfully updated</td></tr>
<tr><td>Failure</td><td>An error occurs and cargo prices do not change</td></tr>
<tr><td colspan="2" rowspan="3">Alternate flows</td><td>1a. No communication between systems
1. The airline pricing engine cannot communicate with the system</td></tr>
<tr><td>2a. No database access
1. The system cannot communicate with the database to save the new data
2. The system notifies an airport administrator and an airline manager of the error
3. Return to step 1</td></tr>
<tr><td>2a.2a. No communication with airport administrator or airline manager
3a. No communication with airport administrator or airline manager
1. The system cannot notify an airport administrator and/or an airline manager
2. Return to step 1</td></tr>
</table>

Requirement ID	Requirement name
FR13	Cargo shipment pricing management

NFR6	Scalability
NFR8	Interlopability

- ### UC21 - Assign flight crews to flights

Use case name		Assign flight crews to flights
Id		UC-21-AssignFlightCrew
Description		An airline manager assigns a flight crew to a scheduled flight
Primary actors		Airline manager
Secondary actors		Airport admin, Flight crew
Preconditions		1. The system has received a schedule for flights 2. The airline manager can access the platform 3. The airline manager is on their dashboard/personal area
Trigger		The airline manager wants to assign flight crews to flights
Main flow		1. The airline manager clicks on the “Assign flight crews” button from their dashboard 2. The system accesses the database and retrieves information about scheduled flights 3. The system shows the airline manager the schedule 4. The airline manager selects a specific flight from the schedule and assigns a specific flight crew to the flight 5. The system saves the updated information in the database 6. The system notifies the airline manager and an airport administrator that the assignment has been done correctly 7. The system notifies the flight crew of their new flight
Postconditions	Success	The flight crew gets assigned to a scheduled flight

		and everyone involved gets notified
	Failure	An error occurs and the system cannot save the assigned flight crew to the scheduled flight
Alternate flows		2a. No database access 5a. No database access 1. The system cannot communicate with the database 2. The system displays an error message and prompts the airline manager to try again 3. Return to step 1
		6a. No communication with the airline manager and/or an airport admin 7a. No communication with the flight crew 1. The system cannot notify the target 2. The system deletes the updated information 3. The system displays an error message and prompts to try again 4. Return to step 4

Requirement ID	Requirement name
FR16	Crew assignment
NFR6	Scalability
NFR8	Interlopability

- ### UC22 - Gate assignment

Use case name	Gate assignment
Id	UC-22-GateAssignment
Description	A ground manager assigns a gate to a scheduled departing or arriving flight
Primary actors	Ground manager
Secondary actors	none
Preconditions	1. The flight is on the schedule

		2. The system has access to the database 3. The user is authenticated and has ground manager account permissions 4. The user is on the dashboard/personal are
Trigger		A ground manager assigns a terminal and gate to a scheduled flight
Main flow		1. The ground manager accesses a “Flight schedule” section 2. The system accesses the database and shows the list of scheduled flights 3. The ground manager clicks on the “Assign gate” button for a specific flight 4. The system shows the gates available and asks the ground manager to assign one to the flight 5. The ground manager selects a suitable gate 6. The system updates the database with the gate assignment 7. The system displays a success message to the ground manager
Postconditions	Success	The system successfully updated the scheduled flight’s assigned gate specified by the ground manager
	Failure	An error occurs and the gate is not assigned to the flight
Alternate flows		2a. No database access 6a. No database access 1. The system cannot communicate with the database to read or write the relevant information 2. The system displays an error message and prompts the ground manager to try again 3. Return to step 1

Requirement ID	Requirement name
FR17	Gate assignment
NFR4	Real-time updates
NFR5	Performance

NFR6	Scalability

- ## UC23 - Schedule management

Use case name	Schedule Management
Id	UC-23-ScheduleManagement
Description	The airport administrator checks the flight schedule and updates operational information when there is a delay, cancellation, or another change. This keeps the schedule clear for airport staff and connected services.
Primary actors	Airport administrator
Secondary actors	Ground manager, Airline flight scheduling engine, Airline manager
Preconditions	1. The flight schedule is saved on the database 2. The user is authenticated and has airport administrator account permissions 3. The user is on the dashboard/personal area
Trigger	The airport administrator wants to change the schedule for a flight
Main flow	1. The airport administrator accesses the “Schedule management” section 2. The system accesses the database and shows the list of scheduled flights 3. The airport administrator clicks on the “Change schedule” button for a specific flight 4. The airport administrator either adds a delay or cancels the departure/arrival 5. The system communicates the schedule change to the airline flight scheduling engine 6. The airline flight scheduling engine creates a new schedule adjusted with the flight delay/cancellation 7. The airline flight scheduling system provides the new updated schedule to the system 8. The system updates the schedule in the

		database 9. The system displays a success message to the airport admin 10. The system notifies the airline managers and ground managers of the schedule changes
Postconditions	Success	The system successfully updates the flight schedule
	Failure	An error occurs and the schedule is not updated
Alternate flows		2a. No database access 8a. No database access 1. The system cannot communicate with the database to read or write the relevant information 2. The system displays an error message and prompts the airport administrator to try again 3. Return to step 1
		5a. No communication between systems 7a. No communication between systems 1. The system cannot communicate properly with the airline flight scheduling engine to give the information about the delay/cancellation or doesn't get a response from the engine 2. The system displays an error message and prompts the airport administrator to try again 3. Return to step 3

Requirement ID	Requirement name
FR15	Flight scheduling
FR18	Landing and takeoff coordination
NFR4	Real-time updates
NFR5	Performance
NFR8	Interlopability

- ## UC24 - Ground staff shift assignment

Use case name		Ground Staff Shift Assignment
Id		UC-24-GroundStaffShiftAssignment
Description		A ground manager assigns shifts to ground staff to ensure all airport needs are covered by the workers
Primary actors		Ground manager
Secondary actors		Ground staff
Preconditions		1. The user is authenticated and has ground manager account permissions 2. The user is on the dashboard/personal area
Trigger		The ground manager needs to create or update work shifts for ground staff
Main flow		1. The ground manager clicks on the “Manage shifts” button 2. The system accesses the database and retrieves any information about shifts if already existent 3. The system shows the ground manager the shift schedule (blank if no data exists yet) 4. The ground manager assigns specific staff members to the shift schedule 5. The system verifies that no conflicts exists 6. The system saves the new or updated information on the database 7. The system notifies the ground staff members involved in the operation of the shift schedule changes 8. The system displays a success message to the ground manager
Postconditions	Success	The shift is assigned and the selected ground staff members can see it in the system
	Failure	An error occurs and the shift schedule doesn’t get updated
Alternate flows		2a. No database access

	6a. No database access 1. The system cannot communicate with the database to read or write the relevant information 2. The system displays an error message and prompts the ground manager to try again 3. Return to step 1
	7a. No communication with ground staff members 1. The system cannot notify the ground staff members of the shift schedule modifications 2. The system displays an error and prompts the ground manager to try again 3. Return to step 1

Requirement ID	Requirement name
FR16	Staff assignment
NFR5	Performance
NFR6	Scalability

- ### UC25 - Create worker accounts

Use case name	Create worker accounts
Id	UC-25-CreateWorkerAccounts
Description	The system administrator creates accounts for workers who need to use the airport system. Each account is created with a specific role so the worker only has access to the parts they need.
Primary actors	System administrators
Secondary actors	Unauthenticated user, Email provider
Preconditions	1. The user must be authenticated and has system administrator permissions

		2. The user is on the dashboard/personal area
Trigger		The system administrator needs to create a new user profile for a member of the airport staff
Main flow		1. The system administrator clicks on the "Create a new account" button 2. The system asks the system administrator for personal information of the new staff member 3. The system administrator inputs all the information needed 4. The system verifies that the nickname and email provided by the system administrator have not been already used in other accounts by checking with the database, and that all fields match the requirements 5. The system generates a suitable temporary password for the new staff account 6. The system creates a new inactive staff member account with all the information provided in the database 7. The administrator enters the worker's information. 8. The system sends a user profile activation link to the new staff member's email through the email provider 9. The new staff member activates the account by pressing the link 10. The system displays a success message to the system admin and the new staff member
Postconditions	Success	The worker account is created and the worker can access the system with the assigned role.
	Failure	An error occurs and no new account is created
Alternate flows		3a. Invalid information 1. The system admin inputs invalid information about the new staff member 2. The system displays an error message and prompts the system admin to try again 3. Return to step 2
		4a. No database access 6a. No database access 1. The system cannot communicate with the database to read or write the relevant

	information 2. The system displays an error message and prompts the system administrator to try again 3. Return to step 2
	8a. No email provider response 1. The system cannot communicate with the email provider 2. The system displays an error message and prompts the user to try again 3. Return to step 2

Requirement ID	Requirement name
FR1	Registration
FR2	User credentials
FR22	Data storing
NFR1	Normative
NFR3	Username uniqueness
NFR8	Interlopability
NFR10	Email uniqueness

- ## UC26 - Update aircraft status

Use case name	Update Aircraft Status
Id	UC-26-Update Aircraft Status
Description	A ground manager changes the status of an aircraft (delayed, cancelled, gate number, departing, landing, etc.)
Primary actors	Ground manager
Secondary actors	none
Preconditions	1. The user is authenticated and has ground

		manager permissions 2. The user is on the dashboard/personal area 3. Database has stored previous flight data
Trigger		The ground manager wants to change the status of an aircraft
Main flow		1. The ground manager clicks the "Flight status" button 2. The system communicates with the database to retrieve information about aircrafts in the airport and shows a list of aircrafts to the ground manager 3. The ground manages clicks the "Update status" button of a specific aircraft 4. The ground manager updates relevant information about the aircraft 5. The system saves the information in the database 6. The system displays a success message
Postconditions	Success	The aircraft status is successfully updated
	Failure	An error occurs and the aircraft status stays the same
Alternate flows		2a. No database access 5a. No database access 1. The system cannot search for flights in the database 2. The system displays an error message and prompts the ground manager to try again 3. Return to step 1

Requirement ID	Requirement name
FR14	Aircraft availability
NFR5	Performance
NFR6	Scalability

- ## UC27 - Update weather information

<table>
<tr><th colspan="2">Use case name</th><th>Update Weather Information</th></tr>
<tr><td colspan="2">Id</td><td>UC-27-UpdateWeatherInformation</td></tr>
<tr><td colspan="2">Description</td><td>The system retrieves updated weather information from an internal weather station placed on airport grounds that collects weather data for close proximities and an external weather information provider that provides data for a larger area</td></tr>
<tr><td colspan="2">Primary actors</td><td>Weather information provider,
In-house weather station</td></tr>
<tr><td colspan="2">Secondary actors</td><td>Airport administrators,
Ground managers</td></tr>
<tr><td colspan="2">Preconditions</td><td>none</td></tr>
<tr><td colspan="2">Trigger</td><td>The system periodically asks the in-house weather station and the weather provider for weather information</td></tr>
<tr><td colspan="2">Main flow</td><td>1. The system requests weather data from both in-house weather station and weather information provider
2. The system saves the data in the database
3. If critical weather data is received, the system notifies airport administrators and ground managers</td></tr>
<tr><td rowspan="2">Postconditions</td><td>Success</td><td>The weather information is updated successfully and can be used by the airport system</td></tr>
<tr><td>Failure</td><td>An error occurs and the system cannot retrieve or save the data provided</td></tr>
<tr><td colspan="2">Alternate flows</td><td>1a. No communication with weather information provider and/or in-house weather station
1. The system doesn’t get any data from the weather information provider and/or the in-house weather station
2. Return to step 1</td></tr>
</table>

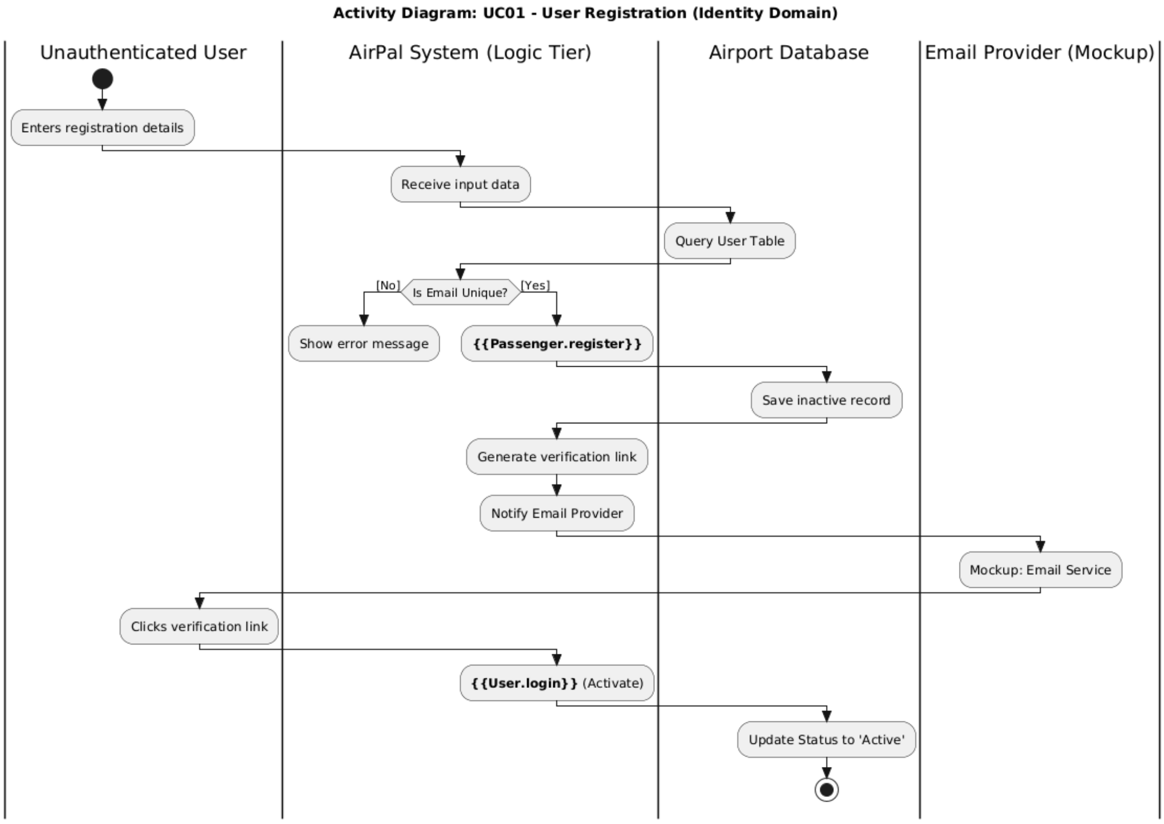
	2a. No database access 1. The system cannot communicate with the database to save the weather information 2. Return to 1
	3a. No communication with airport administrators and/or ground managers 1. The system cannot notify the airport administrators and/or the ground managers of extreme weather conditions 2. Return to step 1

Requirement ID	Requirement name
FR24	Weather information
NFR4	Real-time updates
NFR8	Interlopability

5. Activity Diagrams

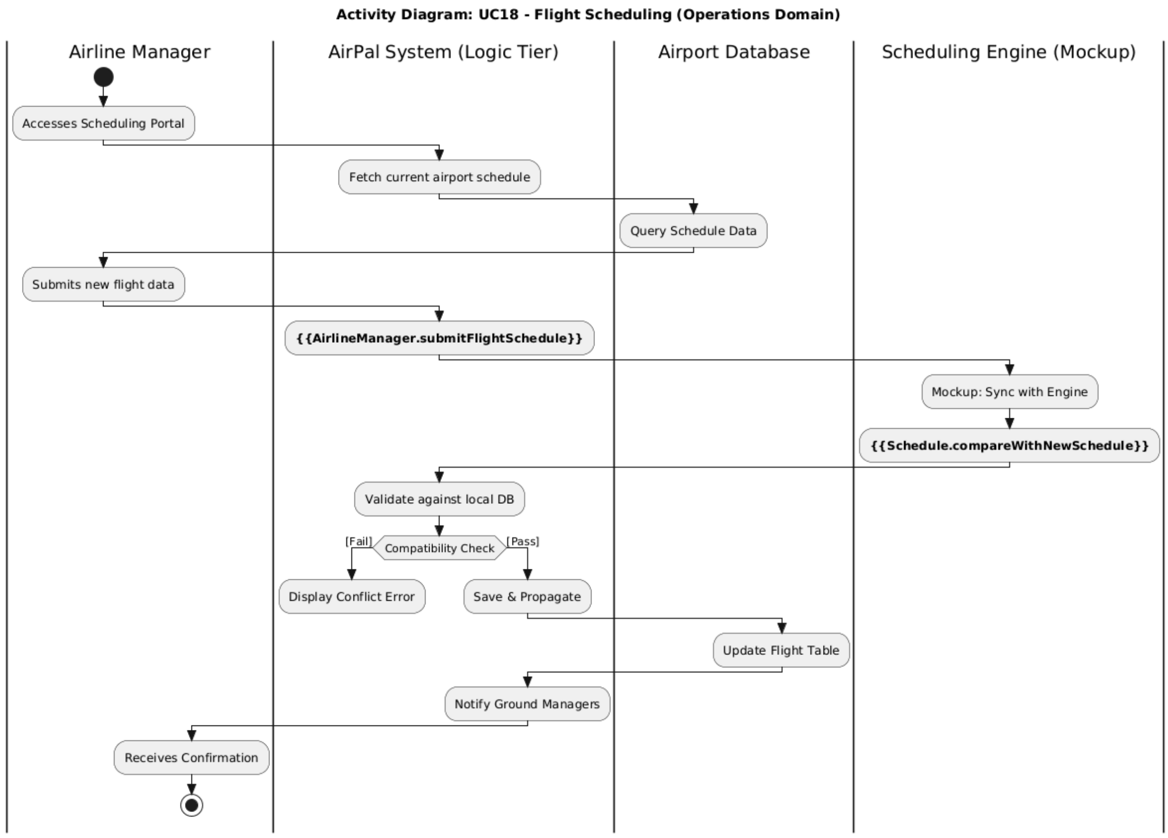
1. Identity Domain: UC01 - User Registration

This diagram captures the dynamic behavior of onboarding a guest and ensures compliance with **NFR10 (Email uniqueness)**.



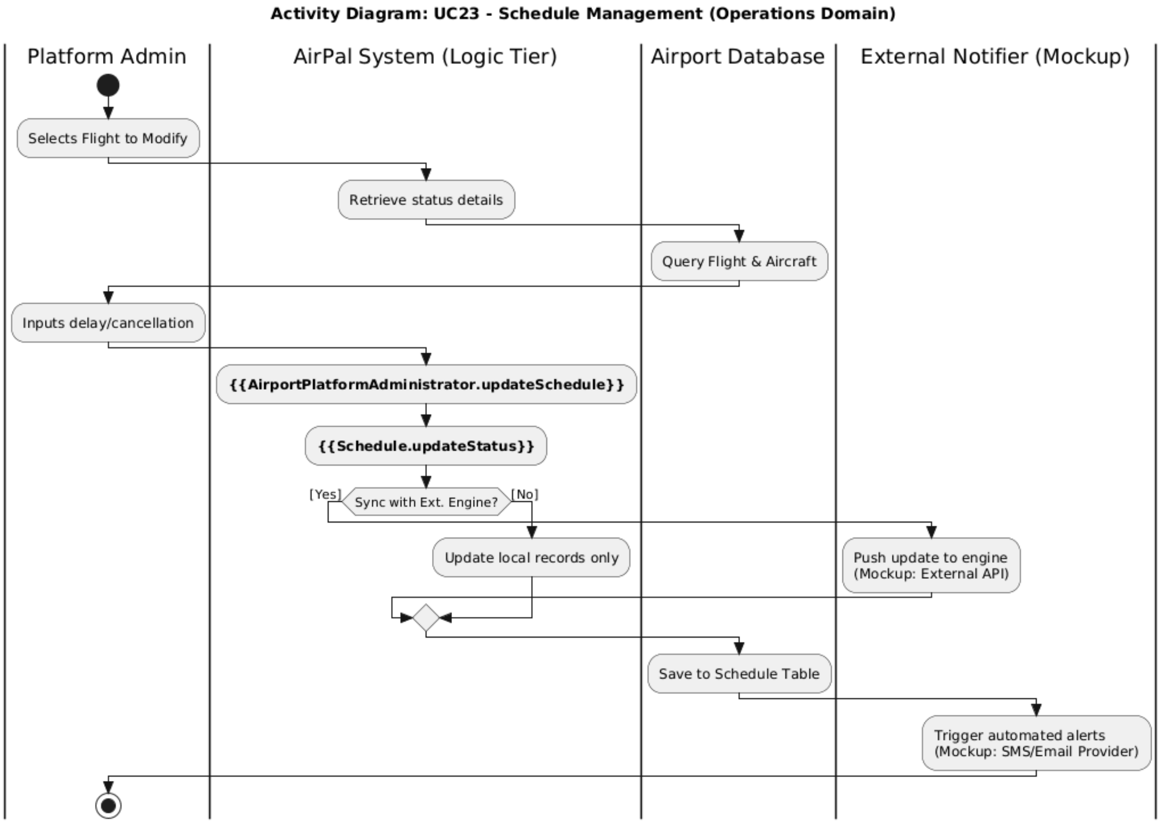
2. Operations Domain: UC18 - Flight Scheduling

This diagram models the "operational brain" logic for how the **Airline Manager** interacts with external engines to update airport data



3. Operations Domain: UC23 - Schedule Management

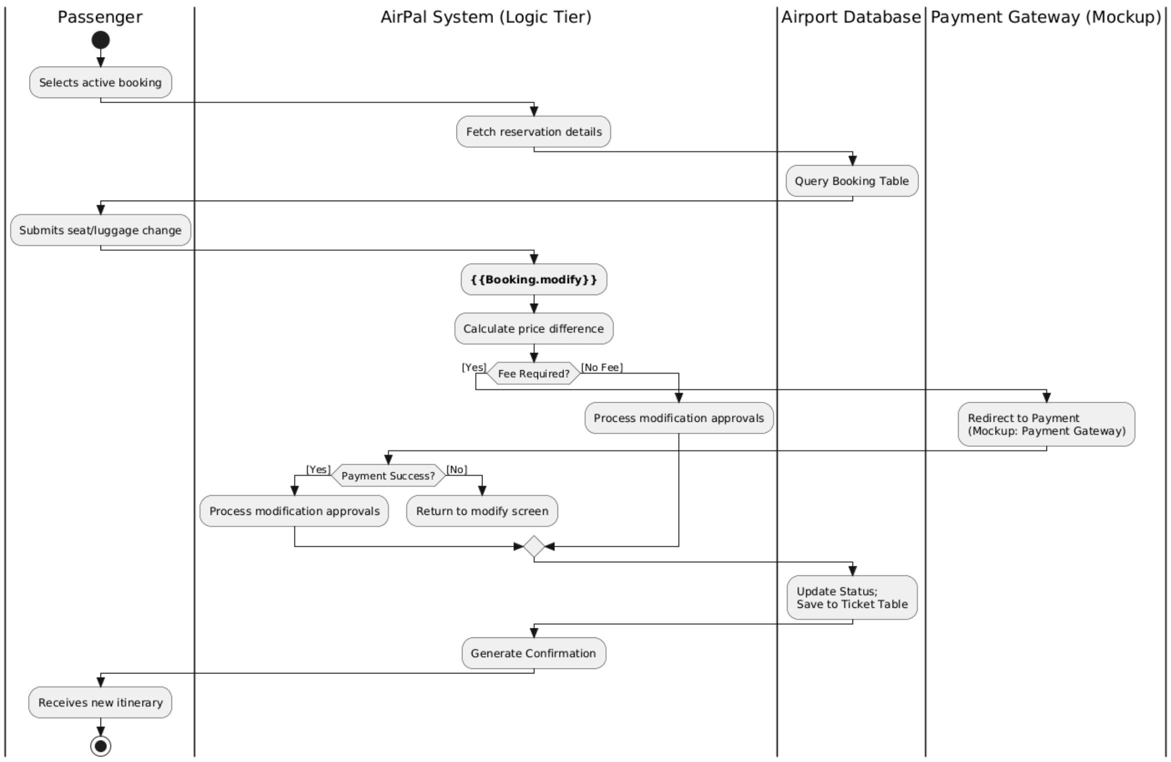
This diagram reflects the **Airport Platform Administrator** role, showing high-level control over delays and cancellations.



4. Commerce Domain: UC08 - Manage Booking

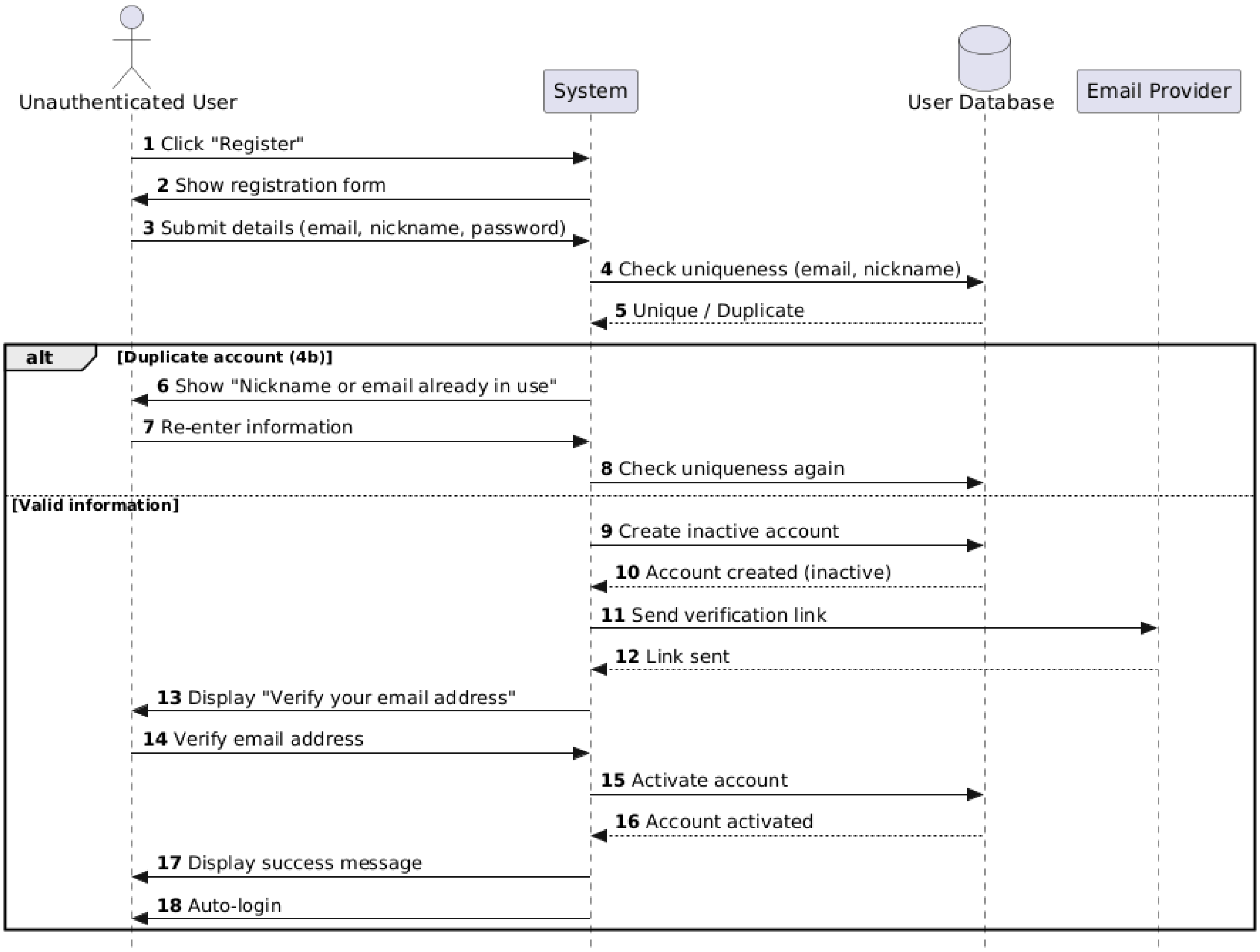
This diagram expands the complex logic for modifying active reservations, including the required external connection for payments.

Activity Diagram: UC08 - Manage Booking (Commerce Domain)

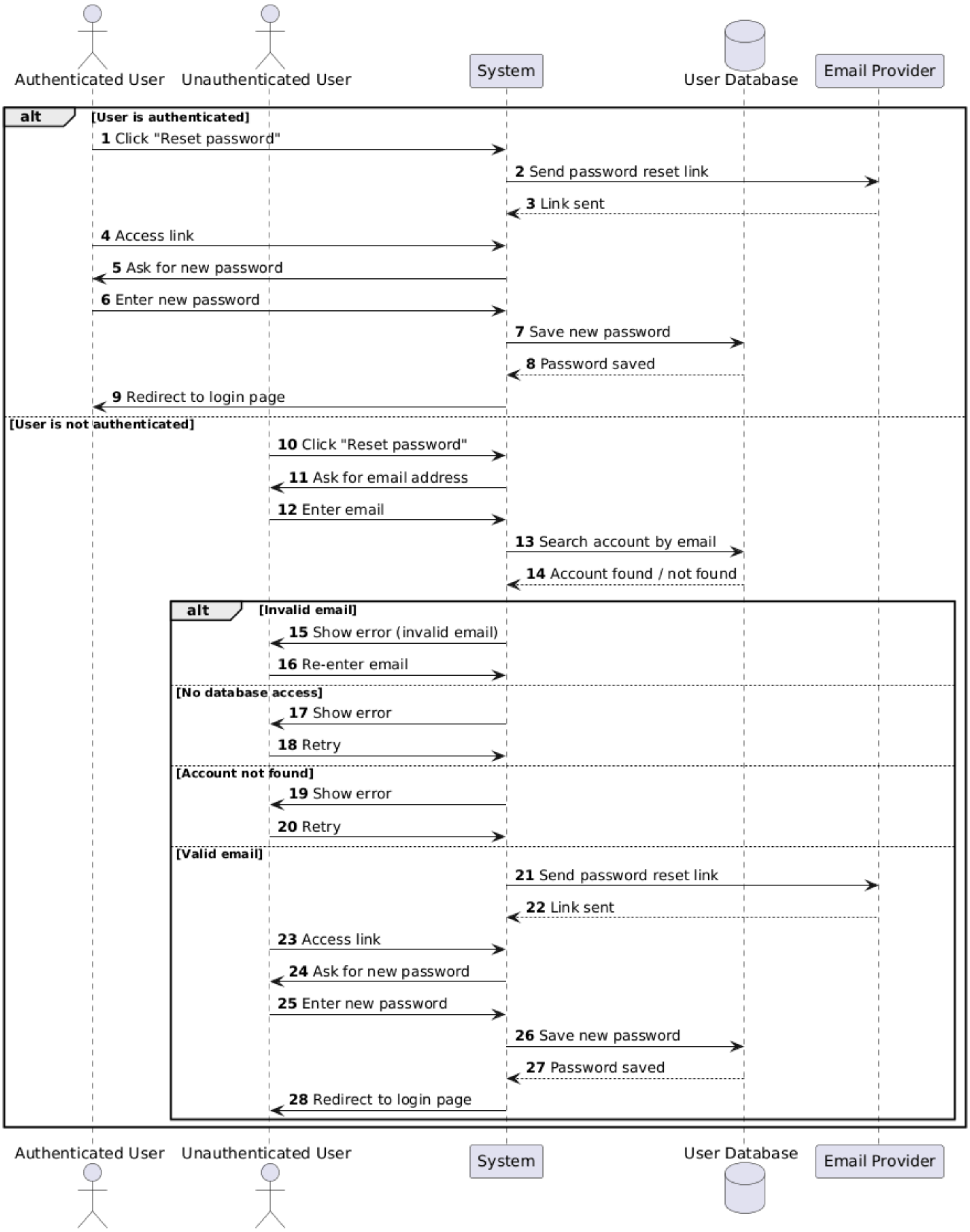


6. Sequence Diagrams

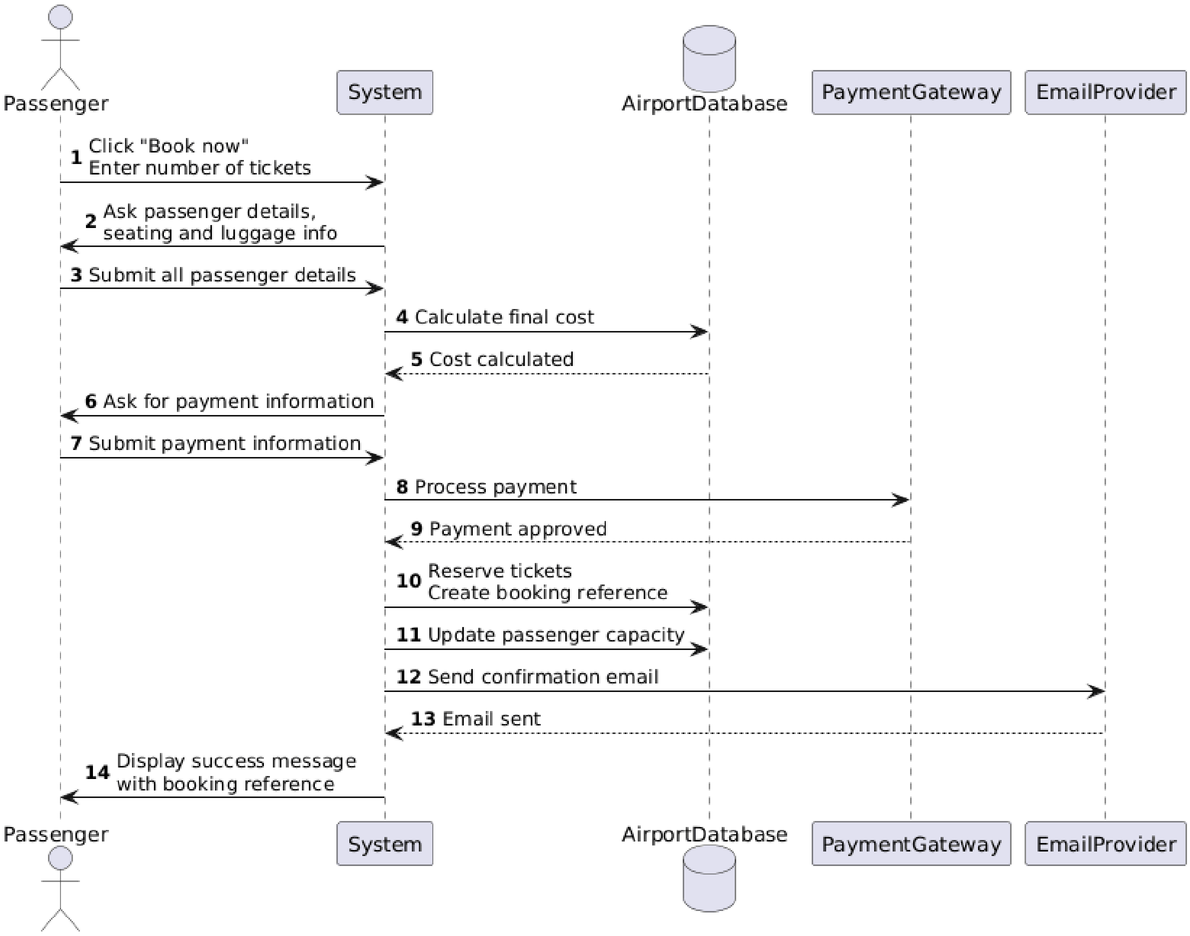
- ## UC01- User Registration



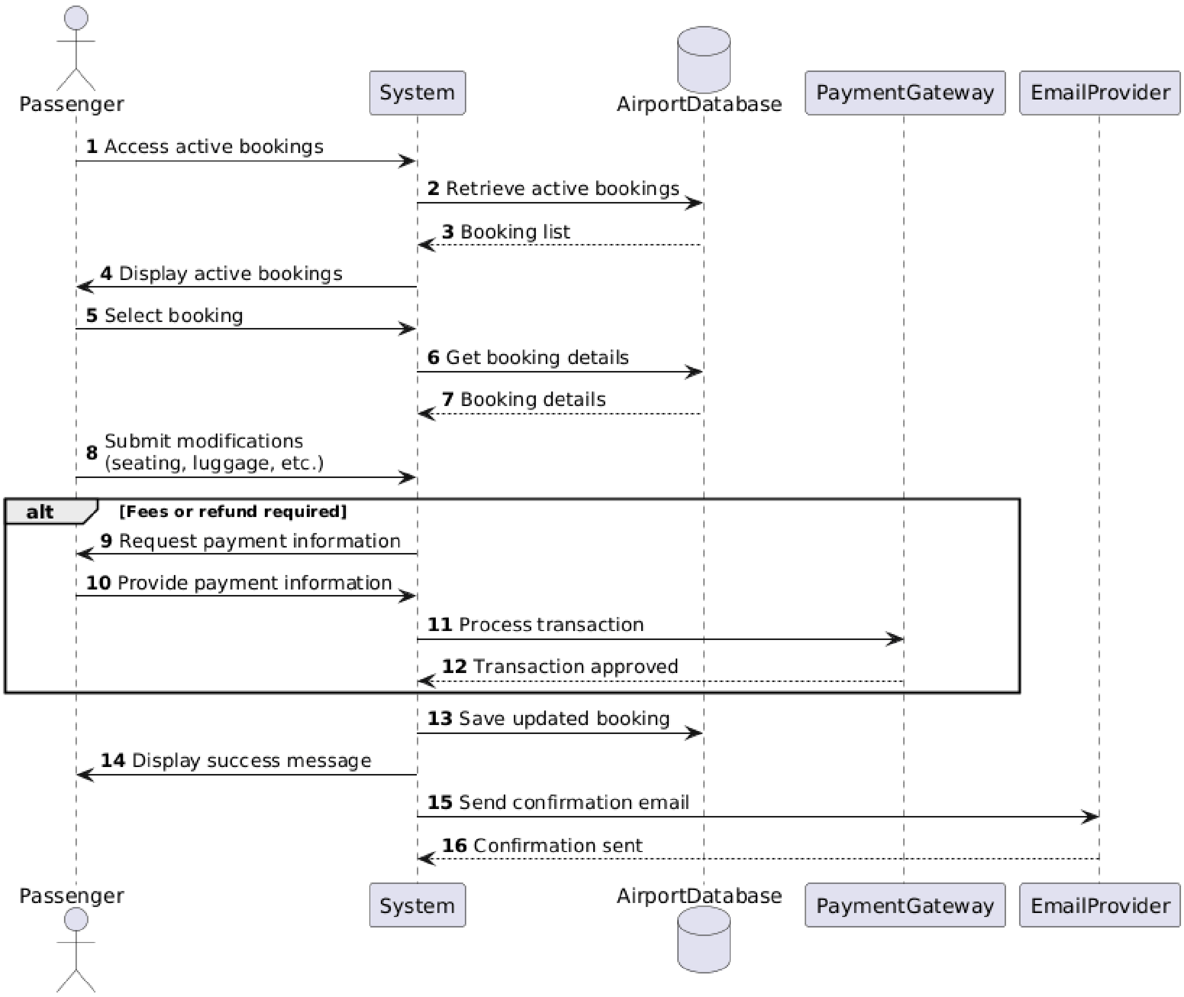
- ## UC04 - Password reset



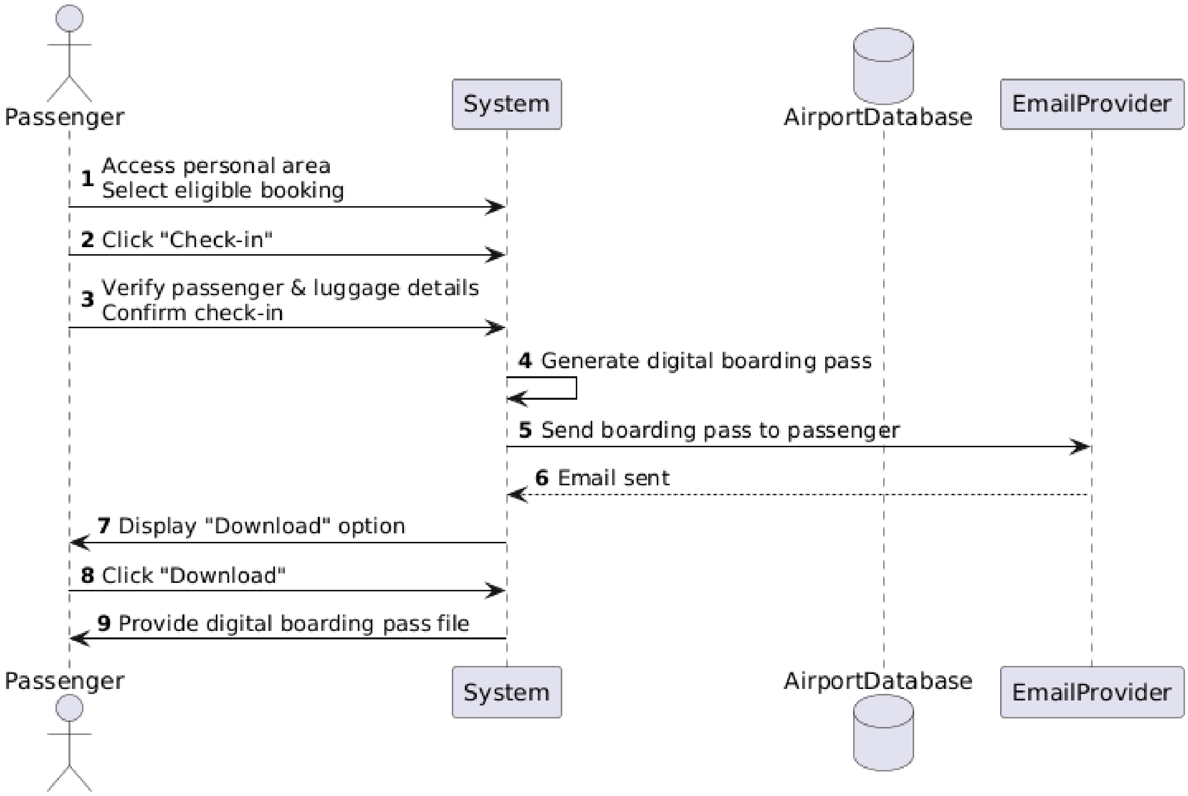
- ## UC07 - Ticket booking



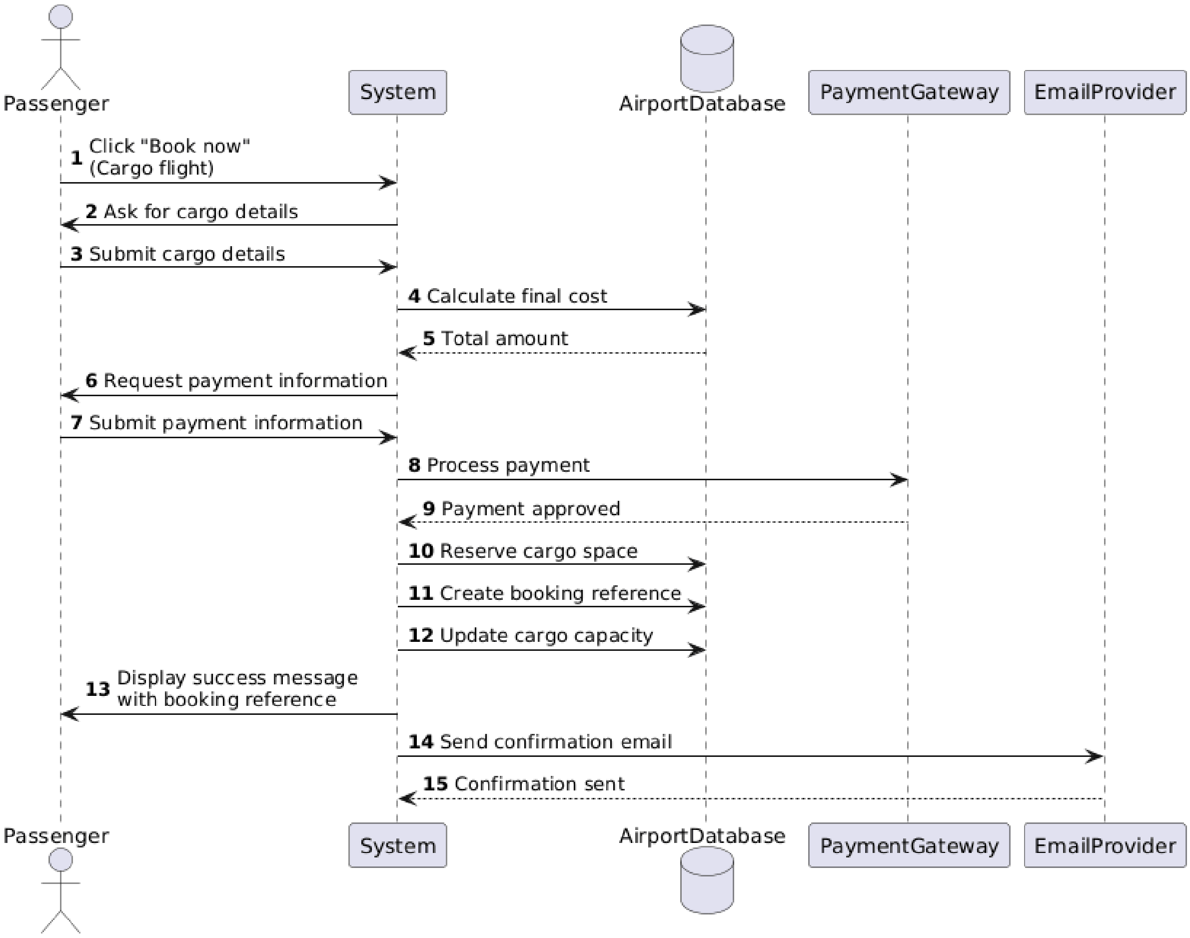
- ## UC08 - Manage booking



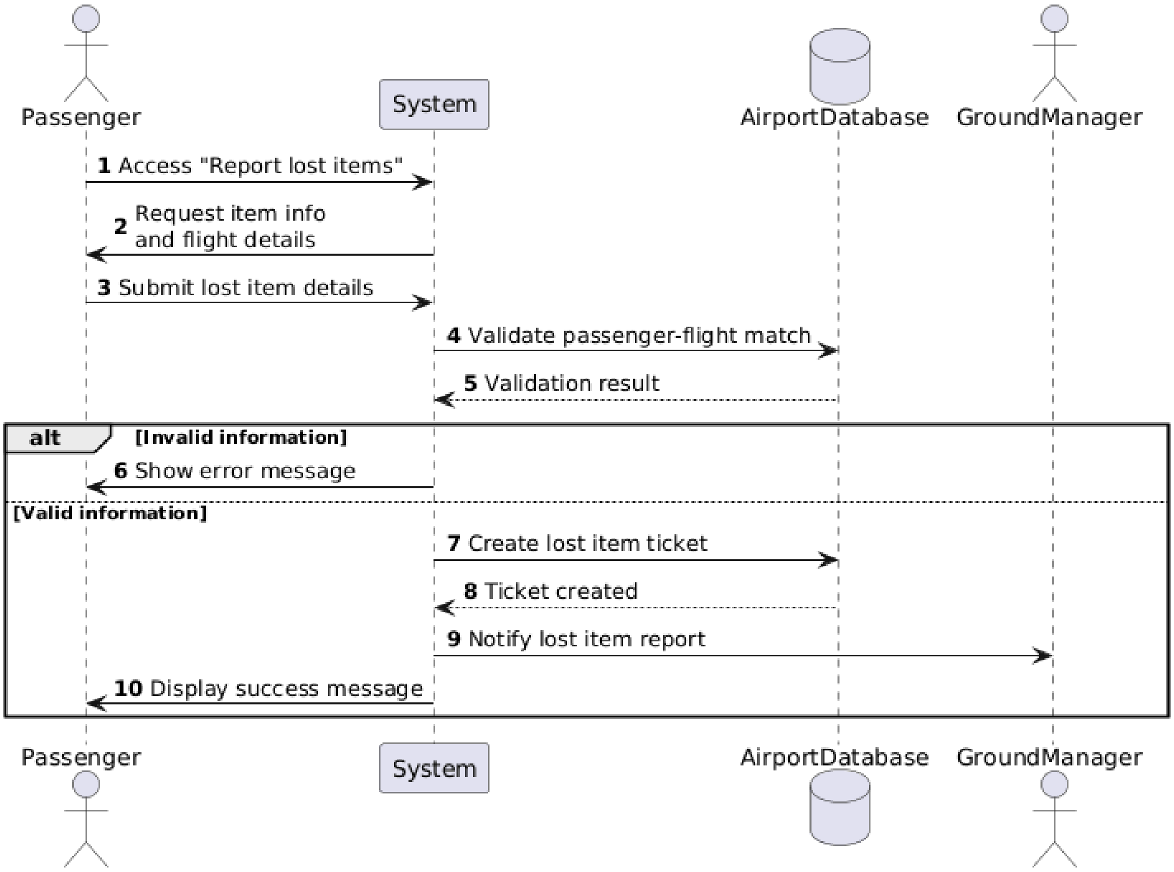
- ## UC09 - Online check-in



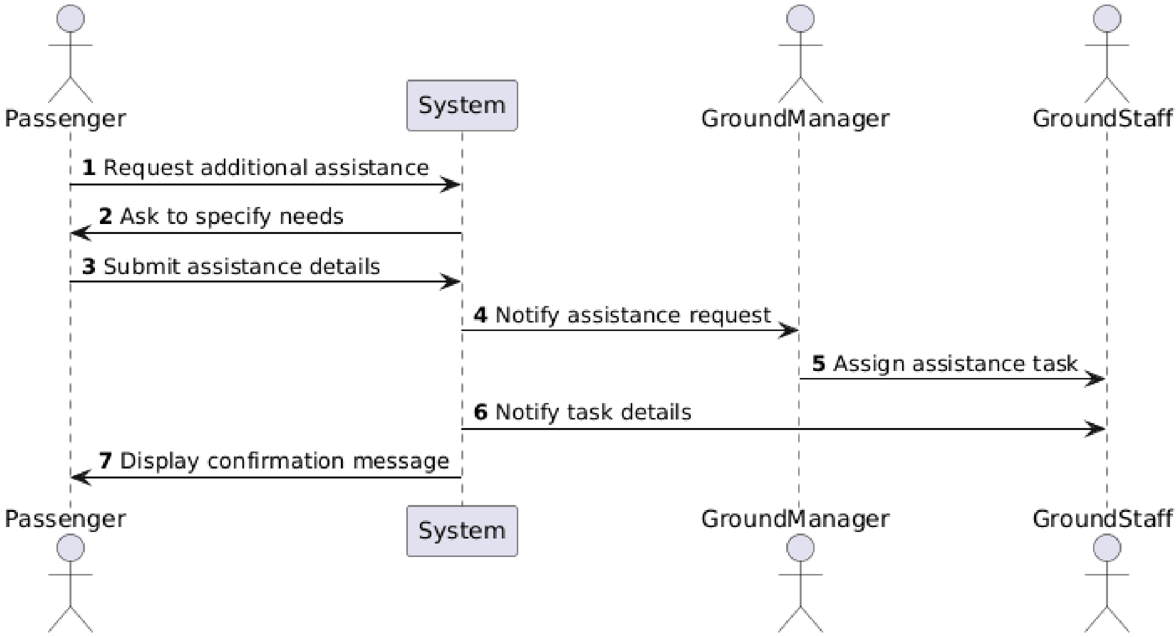
- ## UC13 - Cargo booking



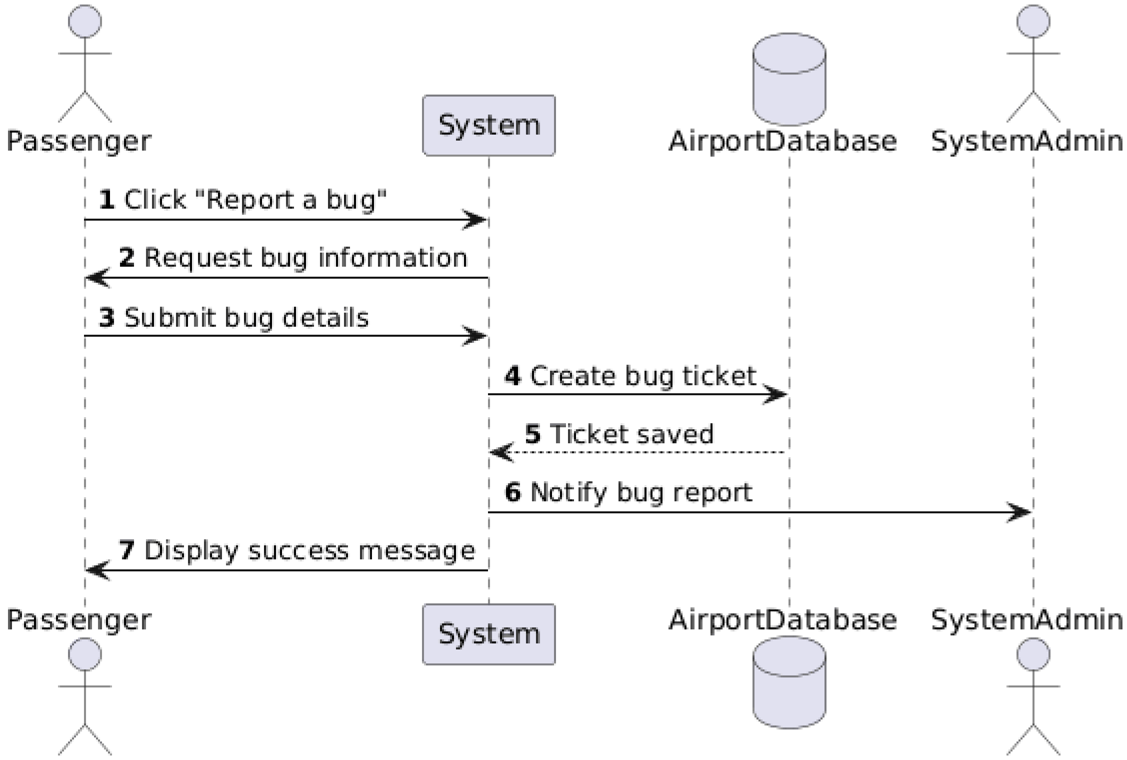
- UC14 - Report lost items



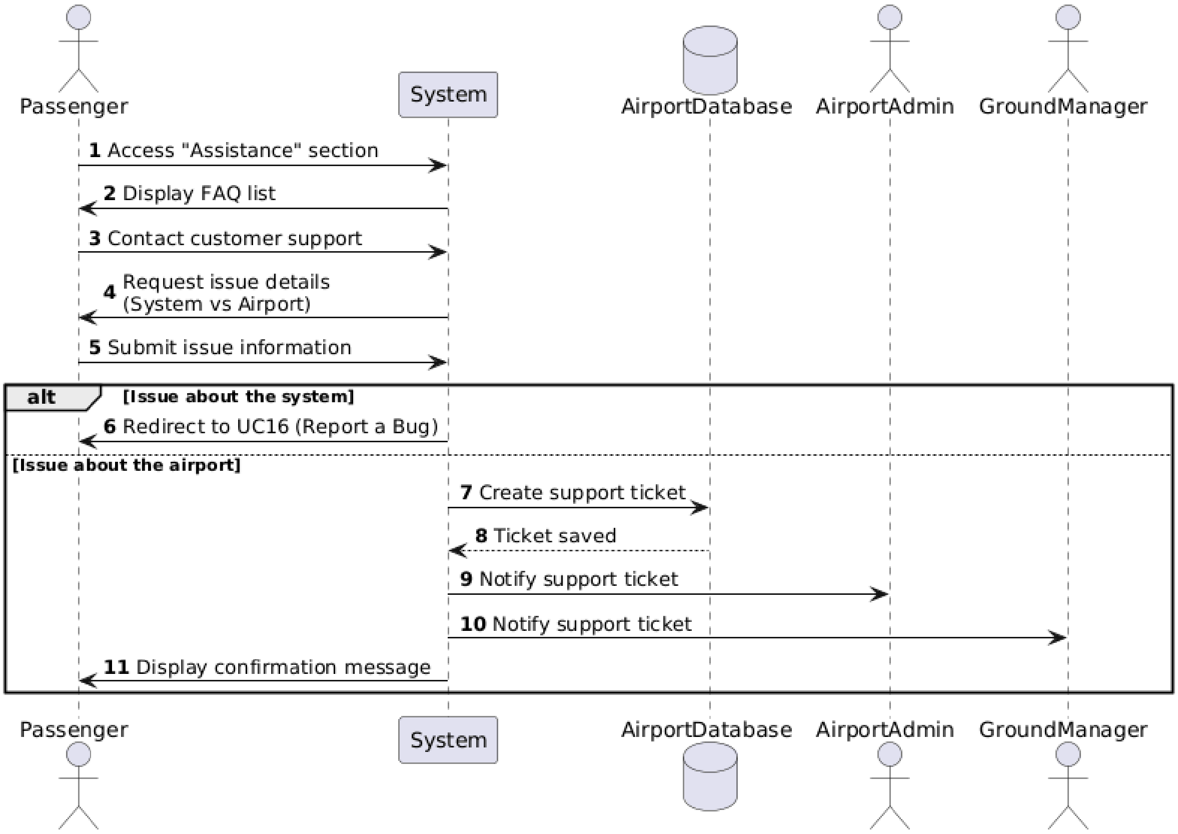
- ### UC15 - Request additional assistance for passengers with disabilities



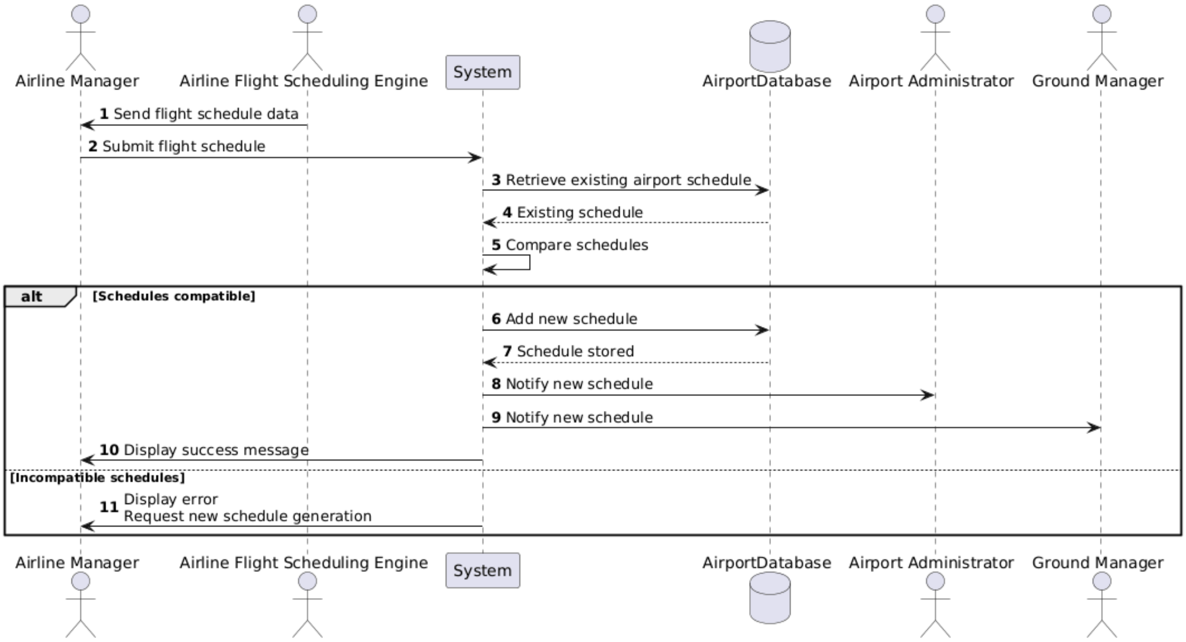
- UC16 - Report a bug



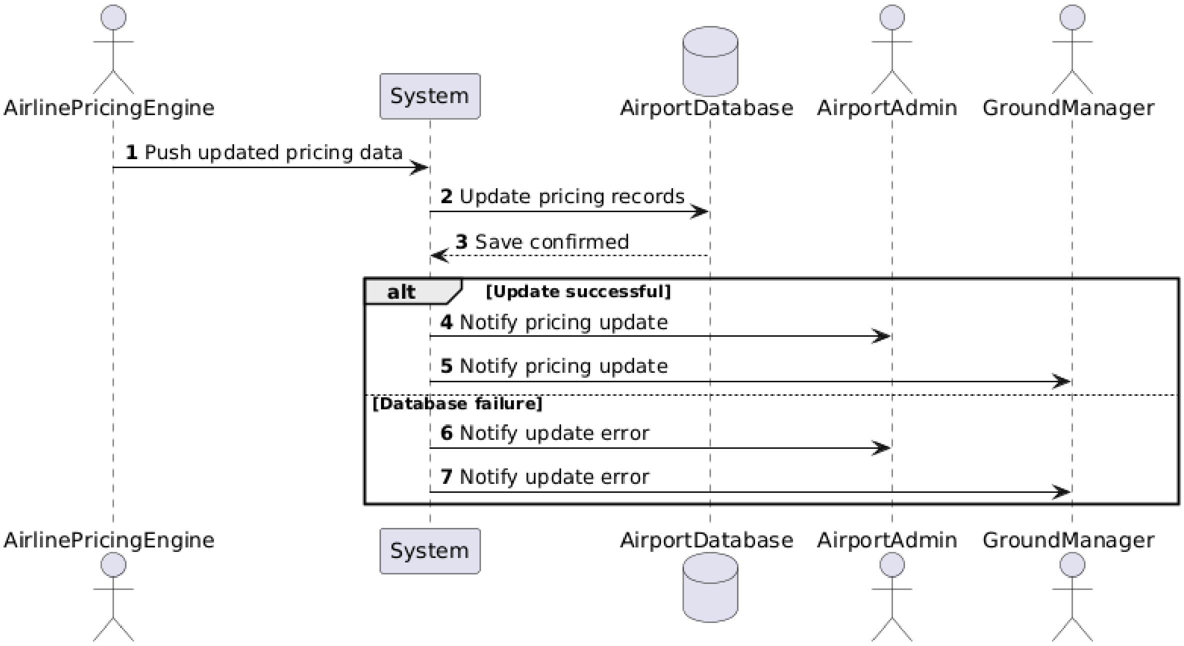
- ## UC17 - Customer support



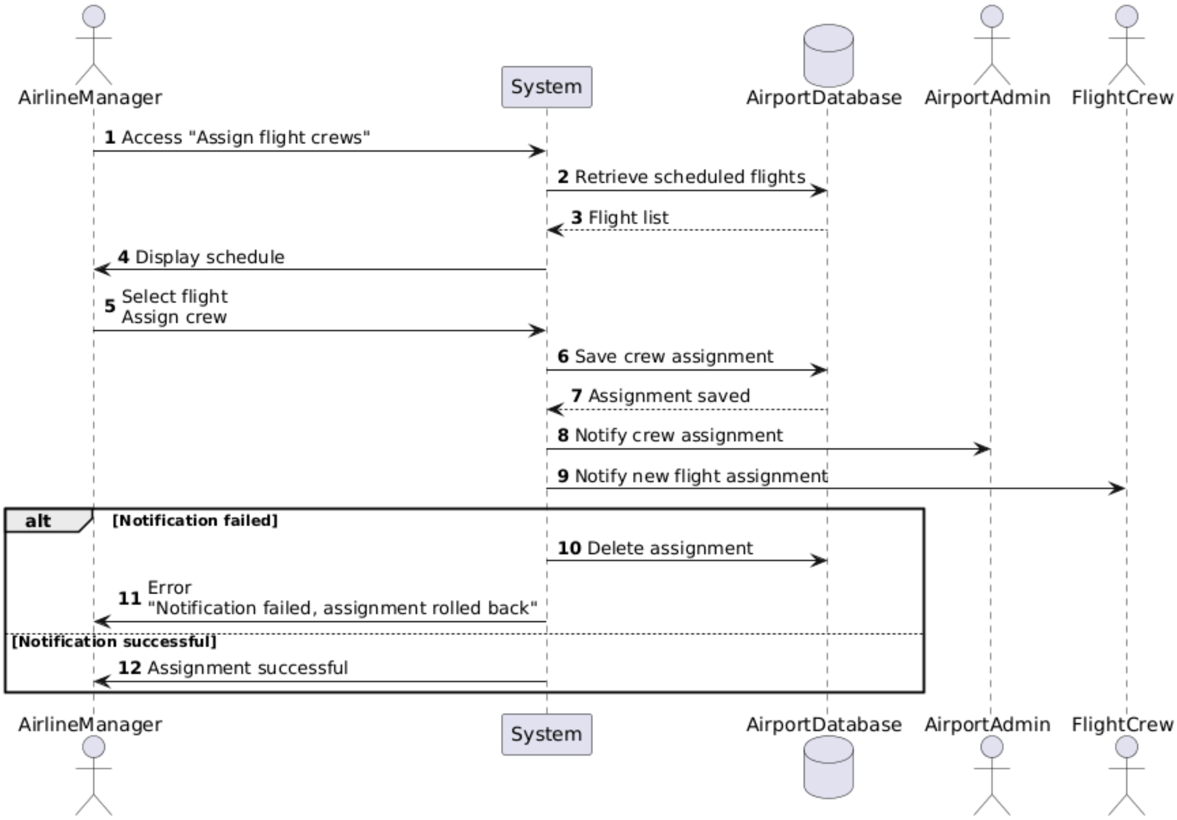
- ## UC18 - Flight scheduling



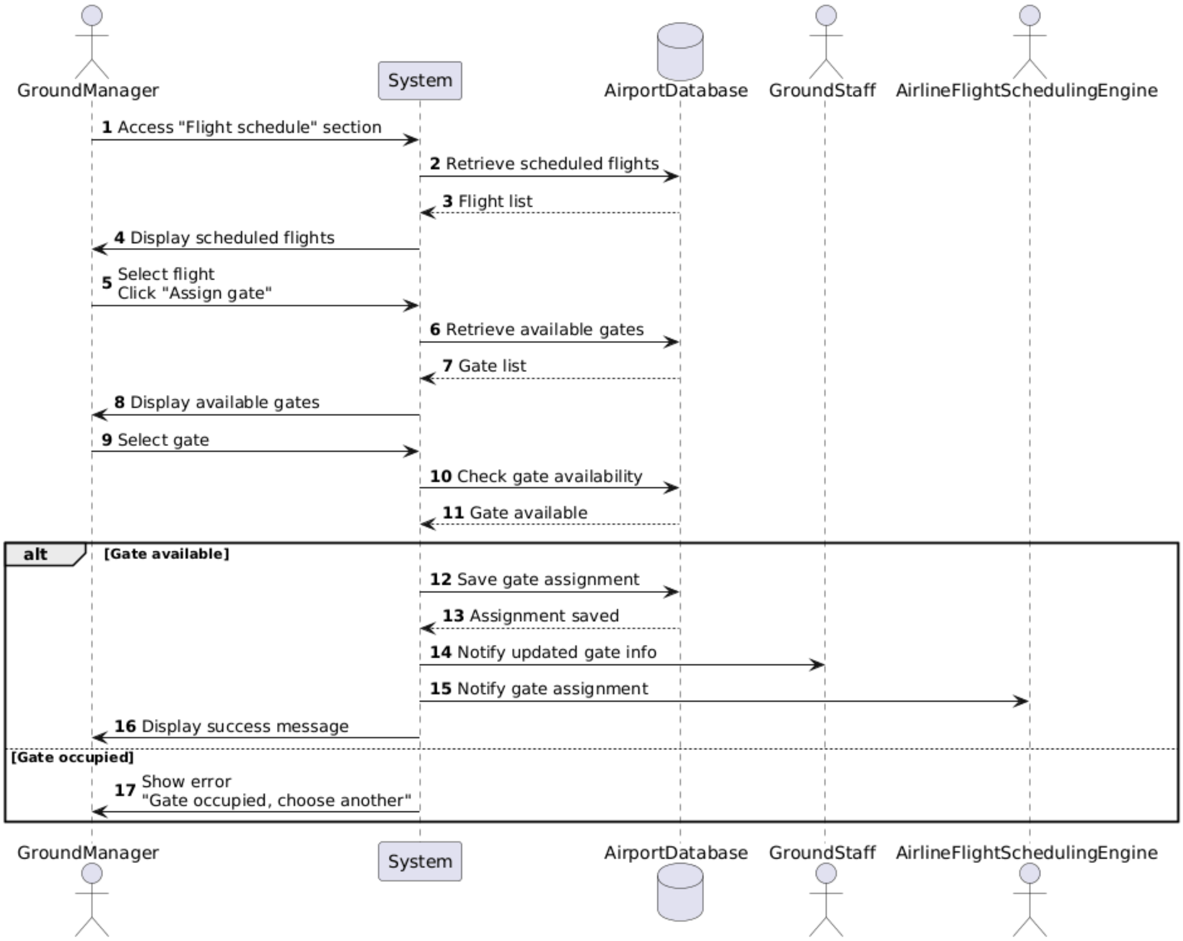
- ## UC19 - Update ticket pricing



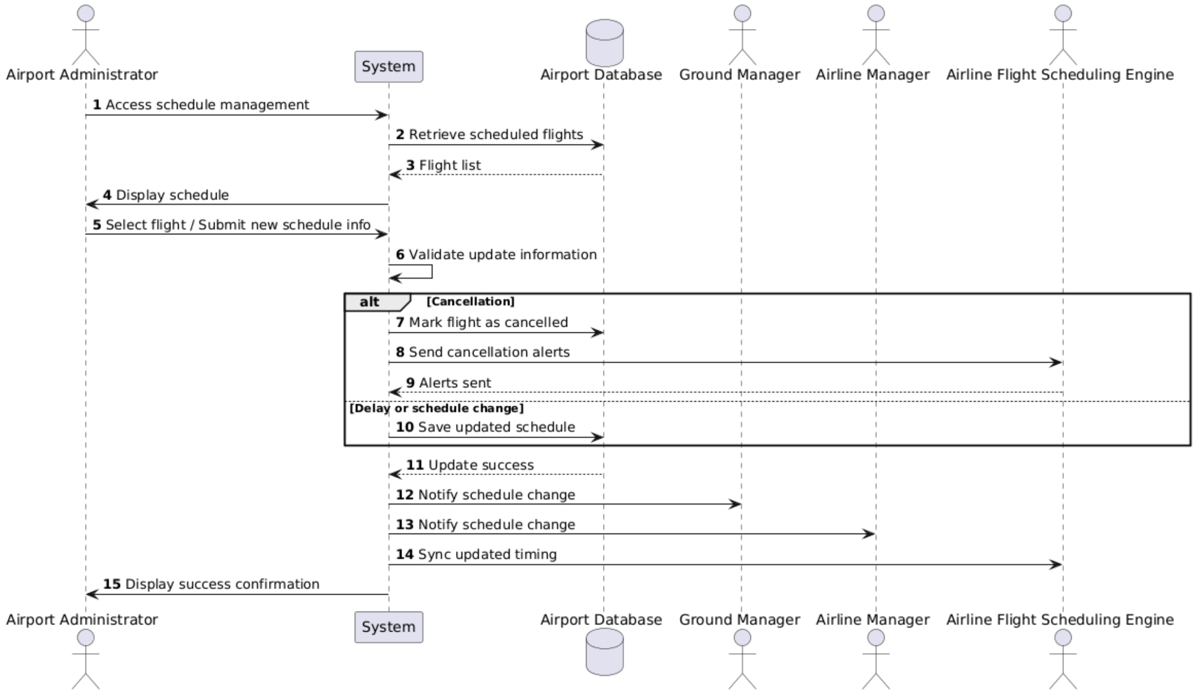
- ## UC21 - Assign flight crews to flights



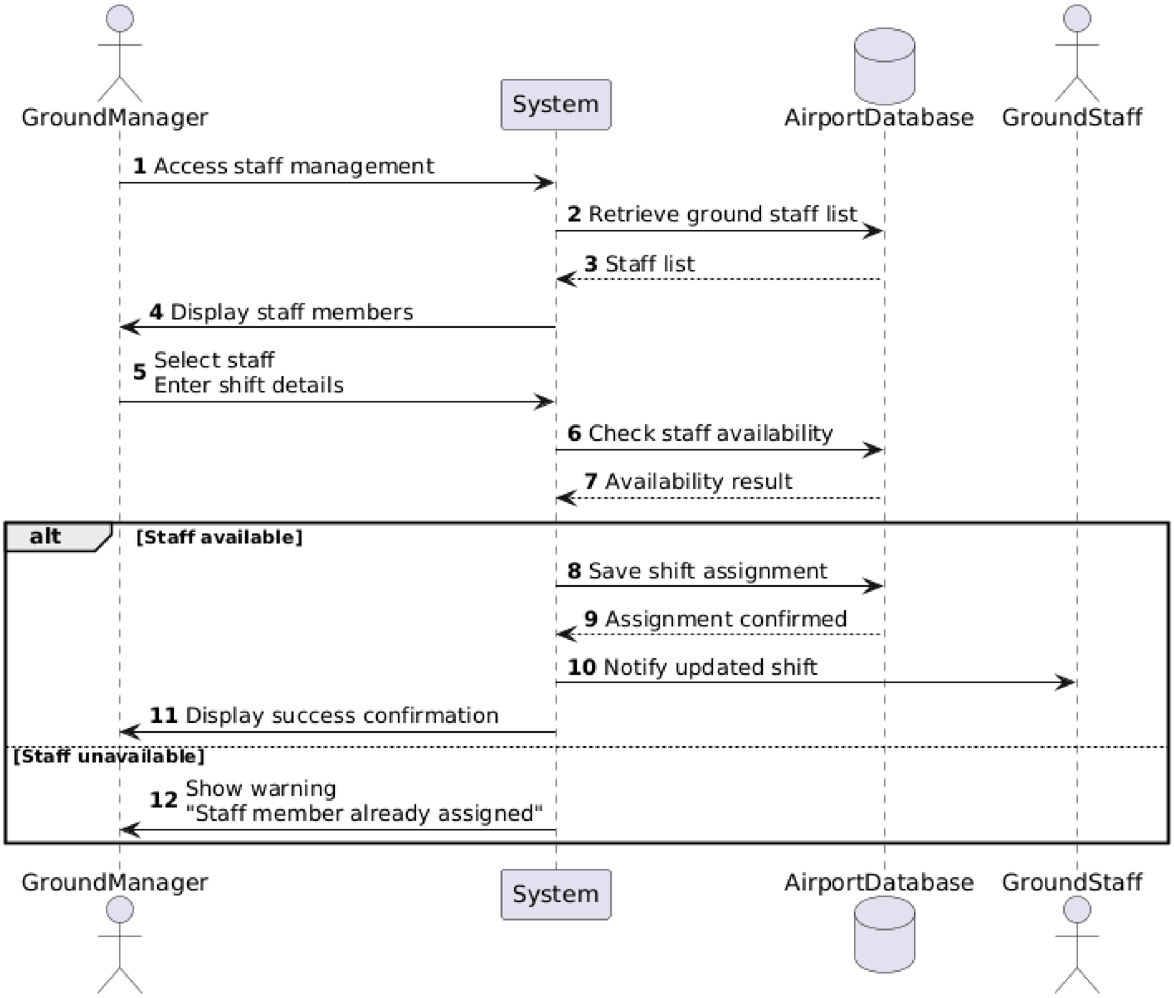
- ## UC22 - Gate assignment



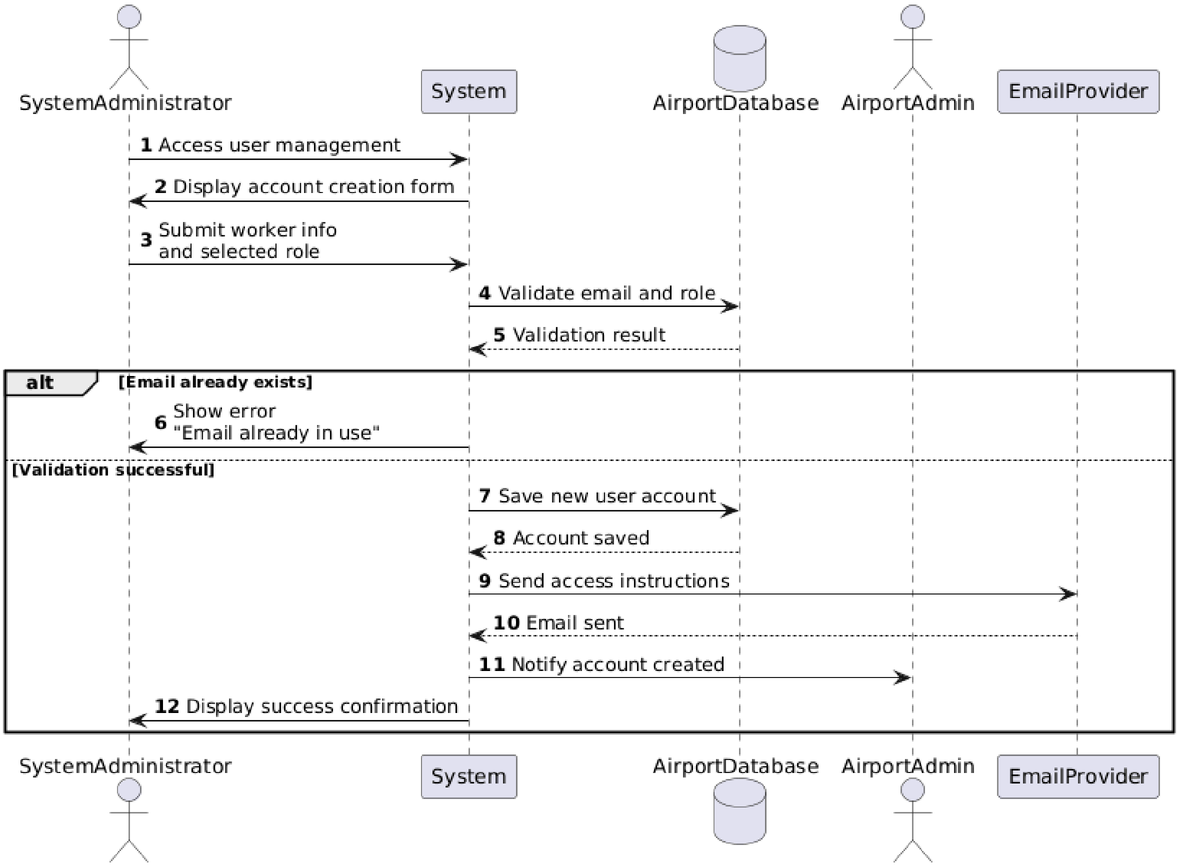
- ## UC23 - Schedule management



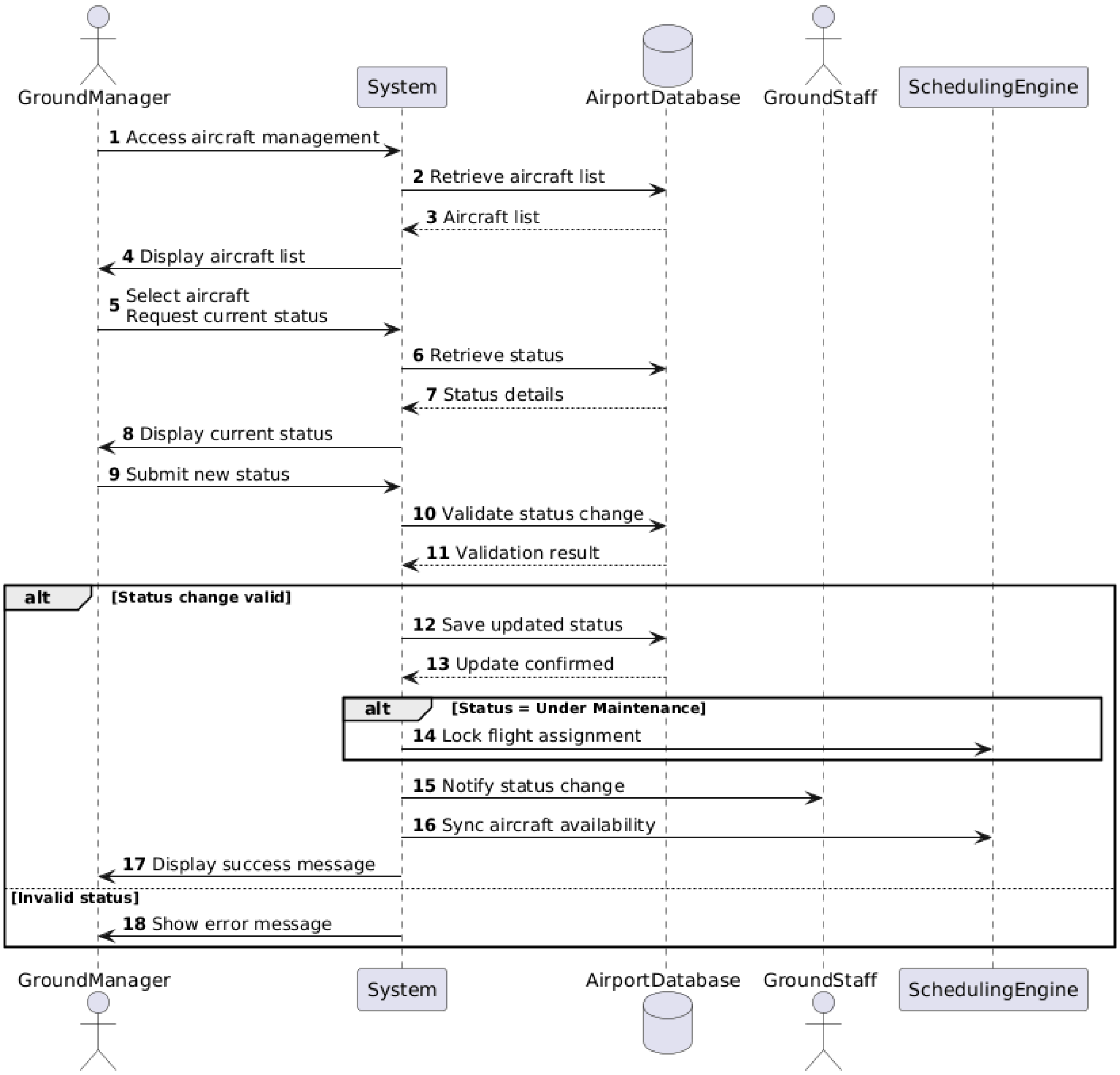
- ## UC24 - Ground staff shift assignment



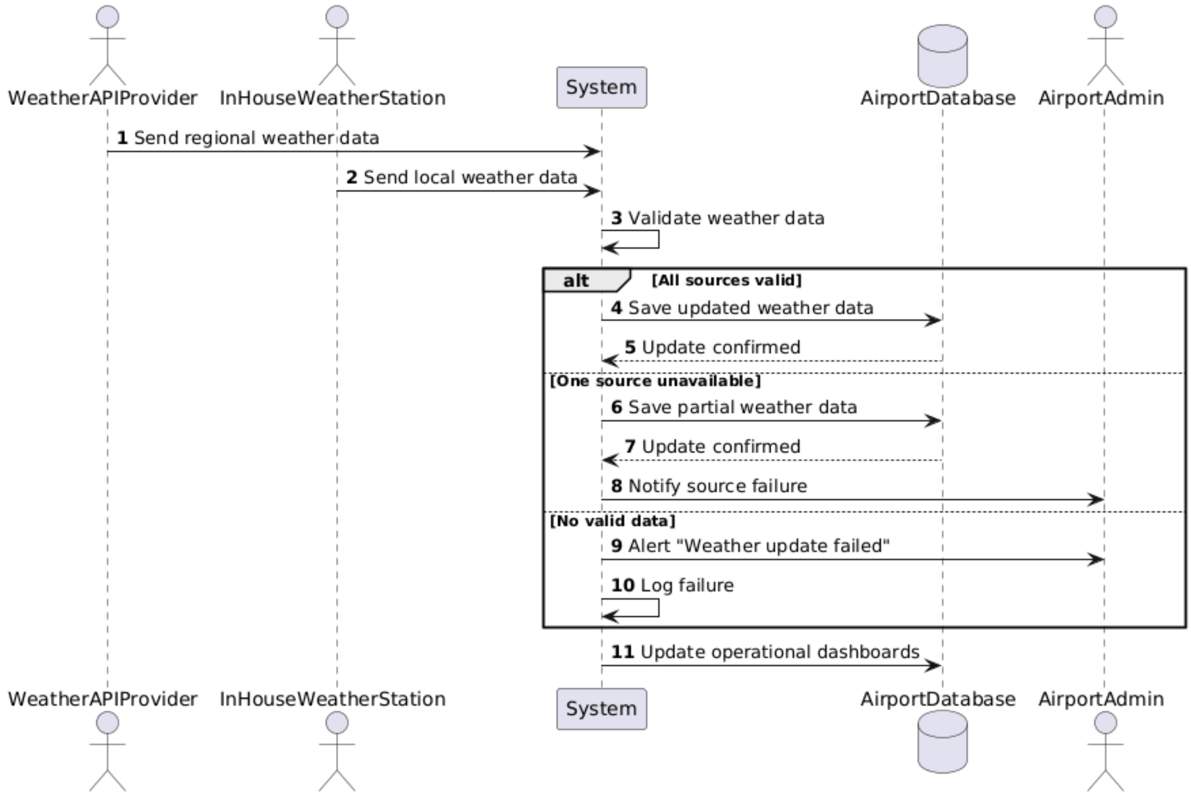
• UC25 - Create worker accounts



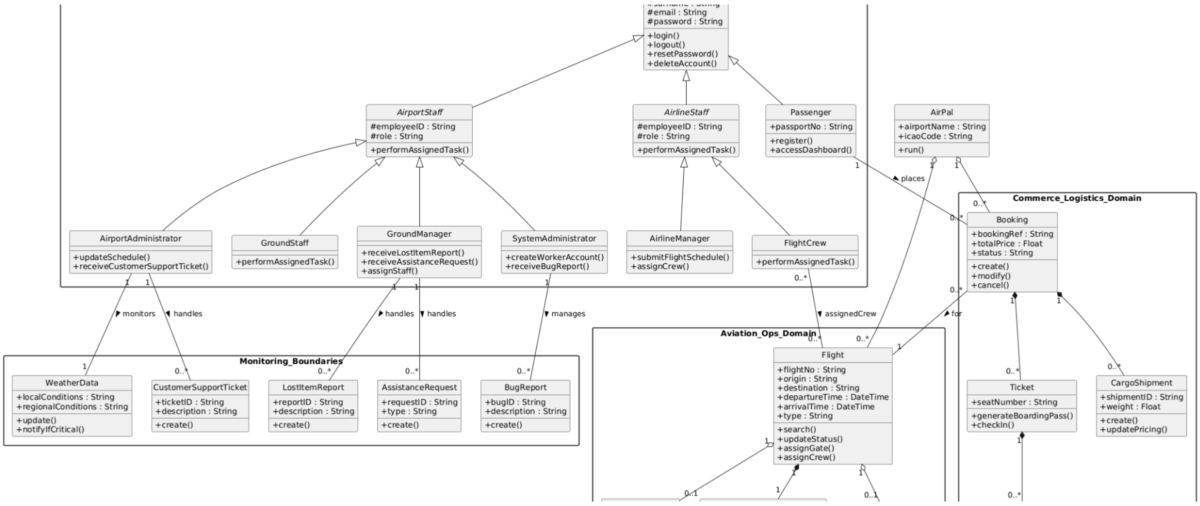
- UC26 - Update aircraft status



- ## UC27 - Update weather information



7. Object-oriented class diagram



[Full size version here]

8. Class Descriptions

8.1 - Identity & Access Management

- ### `User` (abstract)

  Central abstract class representing any authenticated actor in the system. Holds credentials shared by all user types.

Element	Detail
`username`	Unique display name
`name`	First name
`surname`	Last name
`email`	Unique email address used for login and notifications
`password`	Stored as a hash
`login()`	Validates credentials, issues session token (UC02)
`logout()`	Invalidates session (UC03)
`resetPassword()`	Triggers reset email flow (UC04)

`deleteAccount()`	Soft-deletes the account (UC05)

- **`Passenger` (extends `User`)**

Represents a traveller who books flights and uses self-service features.

Element	Detail
`passportNo`	Required for check-in validation
`register()`	Creates a new account with email verification (UC01)
`accessDashboard()`	Loads personal area with bookings and check-in status (UC12)

- **`AirportStaff` (abstract, extends `User`)**

Abstract base for all airport and airline employees. Distinguishes staff from passengers at the model level.

Element	Detail
`employeeID`	Unique staff identifier
`role`	Concrete role label (e.g. “Ground Manager”)
`performAssignedTask()`	Generic entry point for role-specific operations

- **`AirlineStaff` (abstract, extends `User`)**

Abstract base for all employees working for an airline rather than the airport. Kept separate from AirportStaff since airline employees are not airport personnel.

Element	Detail
`employeeID`	Unique staff identifier
`role`	Concrete role label (e.g. “ArlineManager”)

`permorAssignedTAsk()`	Generic entry point for role-specific operations

- ### `SystemAdministrator` (extends `Staff`)

Manages the technical infrastructure and user accounts of the platform.

Element	Detail
`createWorkerAccount()`	Creates accounts for new staff members (UC25)
`receiveBugReport()`	Receives and reviews submitted bug tickets (UC16)

- ### `AirportAdministrator` (extends `AirportStaff`)

Oversees airport-level operations including schedules and customer support.

Element	Detail
`updateShedule()`	Applies changes to the flight schedule (UC23)
`receiveCustomerSupportTicket()`	Handles support issues escalated from passengers (UC17)

- ### GroundManager (extends `Staff`)

Coordinates ground operations and passenger assistance on the airport floor.

Element	Detail
`receiveLostItemReport()`	Receives and processes lost item submissions (UC14)
`receiveAssistanceRequest()`	Manages special assistance needs (UC15)
`assignStaff()`	Assigns ground staff to shifts and tasks (UC24)

- ### AirlineManager (extends `AirlineStaff`)

Represent the airline-side manager responsible for scheduling and crew.

Element	Detail
`submitFlightSchedule()`	Sends a new schedule to the airport system (UC18)
`assignCrew()`	Assigns flight crew members to scheduled flights (UC21)

- ### `FlightCrew` (extends `ArlineStaff`)

Represents flight crew members (pilots, cabin crew) assigned to a specific flight by the Airline Manager.

Element	Detail
`permormAssignedTask()`	Carries out duties for the assigned flight (UC21)

8.2 - Aviation Operations Domain

- ### `Flight`

Core entity representing a single scheduled flight.

Element	Detail
`flightNo`	Unique IATA/ICAO flight identifier
`origin`	Departure airport code
`destination`	Arrival airport code
`departureTime`	Scheduled departure datetime
`arrivalTime`	Scheduled arrival datetime
`type`	Passenger or cargo
`search()`	Supports flight lookup by route and date (UC06)

`updateStatus()`	Updates delay, cancellation, or on-time status (UC26)
`assignGate()`	Links the flight to a gate (UC22)
`assignCrew()`	Links crew members to the flight (UC21)

- ### Schedule

Holds the operational status of a flight's timing plan.

Element	Detail
`status`	E.g. On Time, Delayed, Cancelled
`updatStatus()`	Modifies the current status (UC23)
`compareWithNewSchedule()`	Checks compatibility with an incoming airline schedule (UC18)

- ### Aircraft

Represents a physical aircraft registered in the system.

Element	Detail
`tailNumber`	Unique registration identifier
`updateStatus()`	Marks aircraft as available, in-service or under maintenance (UC26)

- ### Gate

A physical departure or arrival gate at the airport.

Element	Detail
`gateID`	Gate label (e.g. "B06")
`terminal`	Terminal containing this gate
`assignToFlight()`	Associates the gate with a specific

	flight (UC22)

8.3 - Commerce & Logistics Domain

- ### `Booking`

The top-level record of a passenger's reservation. Owns all tickets and cargo shipments made in a single transaction.

Element	Detail
`bookingRef`	Unique reference code sent to the passenger
`totalPrice`	Sum of all tickets and shipments in the booking
`status`	E.g. Confirmed, Modified, Cancelled
`create()`	Initiates a new booking (UC07, UC13)
`modify()`	Updates seats, luggage, or cargo (UC08)
`cancel()`	Cancels the booking with optional refund (UC08)

- ### `Ticket`

Represents one passenger seat within a booking.

Element	Detail
`seatNumber`	Assigned seat on the aircraft
`generateBoardingPass()`	Produces a digital boarding pass (UC09)
`checkIn()`	Marks the passenger as checked in (UC09)

- `CargoShipment`

Represents a cargo booking on a flight.

Element	Detail
`shipmentID`	Unique cargo reference
`weight`	Declared weight
`create()`	Registers a new cargo booking (UC13)
`updatePricing()`	Reflects changes from the airline pricing engine (UC20)

- `Baggage`

Represents a single piece of checked luggage tied to a ticket.

Element	Detail
`tagID`	Unique baggage tag scanned at the carousel
`findCarousel()`	Returns the assigned baggage carousel for a flight (UC11)

8.4 - Monitoring & Boundaries

- `BugReport`

Created when a user reports a technical issue with the system.

Element	Detail
`bugID`	Auto-generated report identifier
`description`	User-submitted description of the issue
`create()`	Submits the report and notifies SystemAdministrator (UC16)

- `CustomerSupportTicket`

Created when a passenger contacts airport support for a non-technical issue.

Element	Detail
`ticketID`	Auto-generated ticket identifier
`description`	Nature of the support request
`create()`	Logs the request and notifies AirportAdministrator (UC17)

- `AssistanceRequest`

Created by a passenger requiring special assistance (e.g. wheelchair).

Element	Detail
`requestID`	Auto-generated request identifier
`type`	Category of assistance required
`create()`	Submits the request and notifies GroundManager (UC15)

- `LostItemReport`

Created when a passenger reports a lost item at the airport.

Element	Detail
`reportID`	Auto-generated report identifier
`description`	User-submitted description of the lost item and flight details
`create()`	Validates passenger-flight match and notifies GroundManager (UC14)

- `WeatherData`

Aggregates weather information from external and in-house sources.

Element	Detail
`localConditions`	Data from the in-house weather station
`regionalConditions`	Data from the external weather API provider
`update()`	Saves new weather data from either source (UC27)
`nootifyIfCritical()`	Triggers alerts to AirportAdministrator when conditions are severe (UC27)

8.5 - Root class

- `AirPal`

Element	Detail
`airportName`	Name of the airport running the system
`icaoCode`	Airport ICAO code
`run()`	System initialization

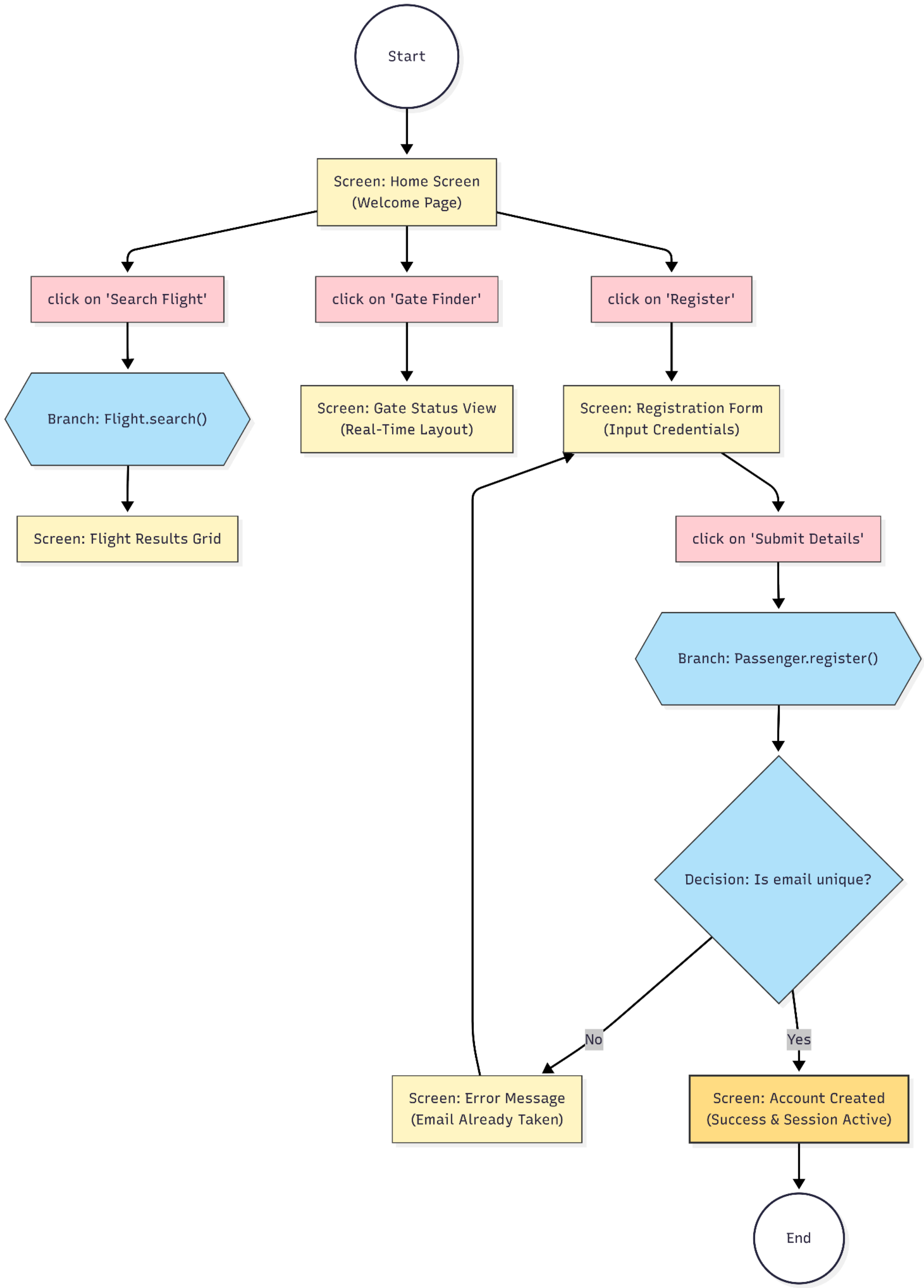
9. Architecture choice and language used

- Architecture choice
- Language used
- Project Dependencies
- Project Data

10. User Flows

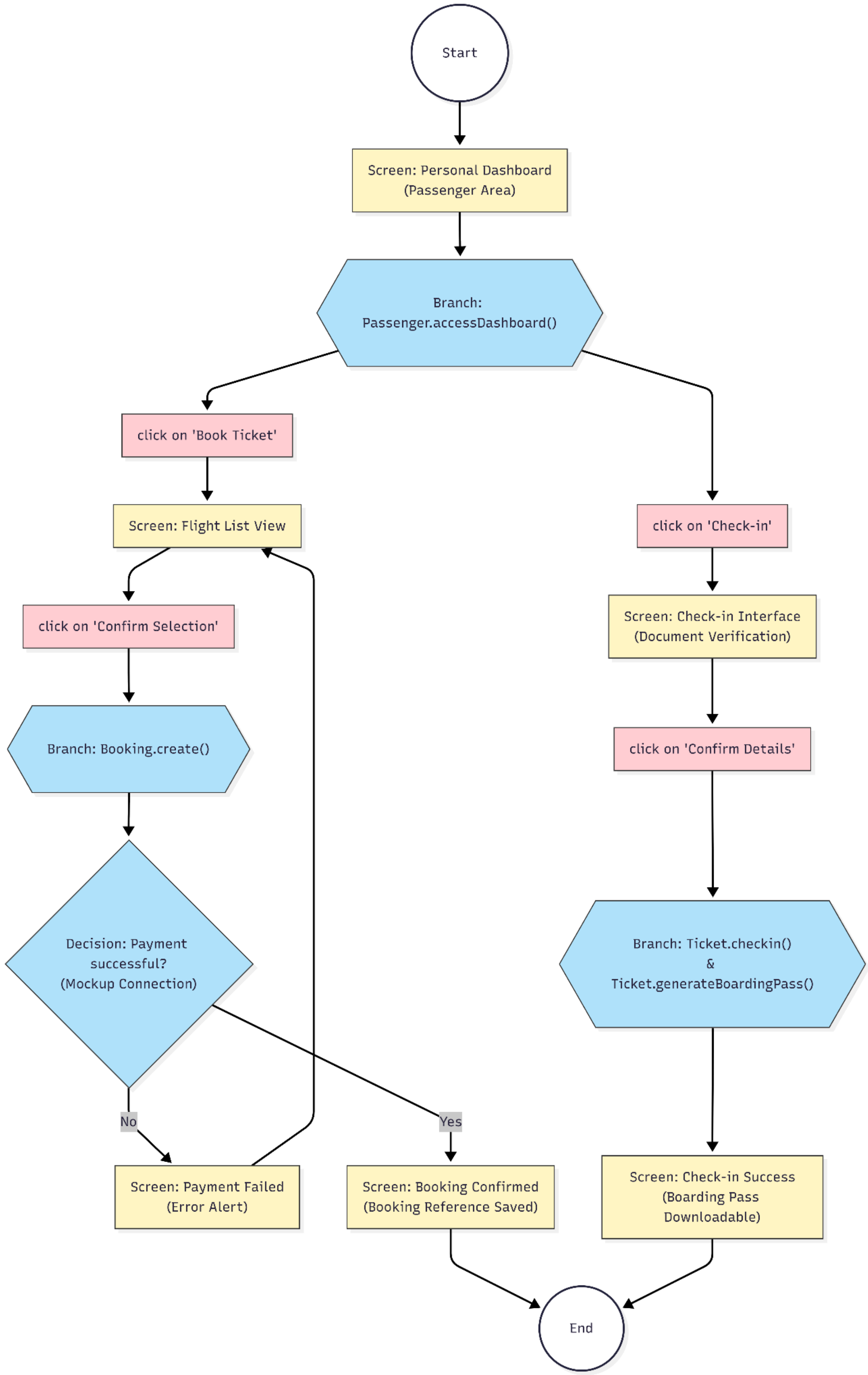
- ## 10.1 Actor: Unauthenticated User

This actor represents a guest who has not yet logged into the system. Their primary objectives are to find flight information and establish a permanent account to access booking services


Start
Screen: Home Screen (Welcome Page)
click on 'Search Flight'
click on 'Gate Finder'
click on 'Register'
Branch: Flight.search()
Screen: Gate Status View (Real-Time Layout)
Screen: Registration Form (Input Credentials)
Screen: Flight Results Grid
click on 'Submit Details'
Branch: Passenger.register()
Decision: Is email unique?
No
Yes
Screen: Error Message (Email Already Taken)
Screen: Account Created (Success & Session Active)
End

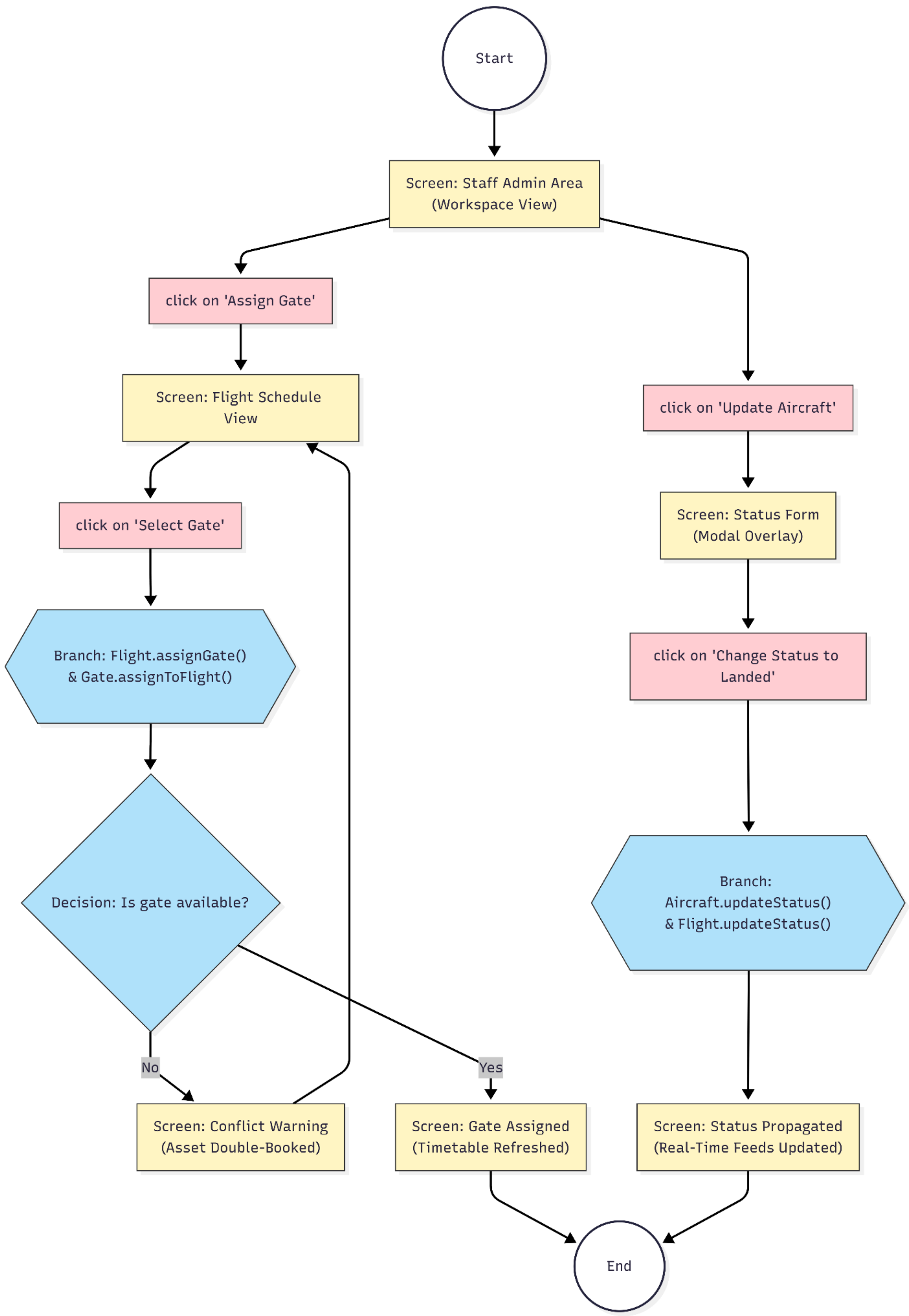
• 10.2 Actor: Authenticated Passenger

A customer who has a valid account. This actor uses the system to manage the "Commerce & Logistics" lifecycle, including booking tickets and completing the check-in process


Start
Screen: Personal Dashboard (Passenger Area)
Branch: Passenger.accessDashboard()
click on 'Book Ticket'
Screen: Flight List View
click on 'Confirm Selection'
Branch: Booking.create()
Decision: Payment successful? (Mockup Connection)
No
Yes
Screen: Payment Failed (Error Alert)
Screen: Booking Confirmed (Booking Reference Saved)
click on 'Check-in'
Screen: Check-in Interface (Document Verification)
click on 'Confirm Details'
Branch: Ticket.checkin() & Ticket.generateBoardingPass()
Screen: Check-in Success (Boarding Pass Downloadable)
End

- ### 10.3 Actor: Ground Manager (Internal Staff)

An internal staff member responsible for "Airport Operations Management." They coordinate real-time logistics such as gate assignments and aircraft readiness


Start
Screen: Staff Admin Area (Workspace View)
click on 'Assign Gate'
Screen: Flight Schedule View
click on 'Select Gate'
Branch: Flight.assignGate() & Gate.assignToFlight()
Decision: Is gate available?
No
Yes
Screen: Conflict Warning (Asset Double-Booked)
Screen: Gate Assigned (Timetable Refreshed)
click on 'Update Aircraft'
Screen: Status Form (Modal Overlay)
click on 'Change Status to Landed'
Branch: Aircraft.updateStatus() & Flight.updateStatus()
Screen: Status Propagated (Real-Time Feeds Updated)
End

- ## 10.4 **Actor: Airport Administrator**

This internal actor manages the operational integrity of the airport. Their primary goals are to update flight schedules during disruptions (delays/cancellations) and resolve customer support tickets escalated from passengers